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FUNDAMENTAL PRINCIPLES OF DATA SCIENCE MASTER'S THESIS

Data-Driven Cognitive Phenotyping in Acquired Brain Injury: A Clustering Analysis with Imputation Robustness Assessment

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Abstract

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MSc

Data-Driven Cognitive Phenotyping in Acquired Brain Injury: A Clustering Analysis with Imputation Robustness Assessment

by Zoltan Kunos

Neuropsychological evaluations of people who have experienced an acquired brain injury (ABI) are often incomplete. This research addressed this by exploring whether the use of "clustering" would allow for a consistent representation of cognitive function despite two challenges: What method(s) would be used to replace missing information from the evaluation? How could the characteristics of cognitive ability be represented?

The historical component of the research included 17,406 assessments of 7,285 patients across six different areas tested using fourteen tests. Ten methods of imputation were used as part of the research. Using UMAP+HDBSCAN, a three band description of overall severity was generated based upon the entire set of data. Across imputations, the fixed-reference method resulted in a mean adjusted Rand index (ARI) of 0.710. While there seemed to be a number of small associations between the cluster groups and the clinical units, there was no clear added patient-grouped predictive value beyond broad diagnostic classifications. These findings did not establish distinct cognitive phenotypes.

In addition, the exploratory portion of the research was completed by taking the first eligible assessment for each individual patient. There were 5,828 discovery patients and 1,457 validation patients. Preprocessing was fitted within the discovery sample prior to conducting clustering within the original data space. In addition to clustering using the original data, the exploratory portion utilized several alternative methods including: Robust mixture models fit to the observed data; Twenty independently refit stochastic imputations; Fifty patient subsamples; Mixed-effects trajectory models.

Two-group k-means clustering was selected in each of the 20 imputation screens. The average pairwise validation ARI for these cluster analyses was 0.959. ARI for three-group k-means was slightly lower (0.935). Changing from robust to equal-domain scaling resulted in large changes in the three-group assignments (ARI = 0.243). Test availability differed across groups. In terms of fit with the observed data, robust t mixtures performed better and preferred an eight-component solution, the upper limit allowed during model selection. Among the 2,013 patients eligible for

longitudinal assessment, multi-class trajectory models introduced groups below the specified 5% size threshold.

Therefore, we have shown there to be evidence supporting replicable grouping of cognitive test results based on certain assumptions regarding representation and imputation. However, it cannot be concluded that these groupings represent separate clinical phenotypes or could be used for treatment purposes. Additional independent measures of outcomes and additional samples from external cohorts will need to occur before clinical use. CogDash presents both analysis designs for research, with the analysis code available at https://github.com/zkunos/thesis_project.

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Statement on the Use of Artificial Intelligence

According to the guidelines provided by the Facultad de Matemàtiques i Informàtica of the Universitat de Barcelona regarding the use of large language models (LLMs) in final master's thesis research I make the following statement.

Throughout my research I utilized Claude (Anthropic) as an AI-assistant tool to improve my research workflow.

For the October 2026 clustering extension, Codex (OpenAI) also assisted with literature assessment, implementation, numerical checks, results synthesis and integration into the report. These contributions do not replace the author's responsibility for reviewing the code, evidence and interpretation.

Selected passages were revised using Walter AI, with Codex assisting in checking the suggestions against the source. The revisions were checked in blocks of no more than 2,000 characters using Walter's detector and, in a subsequent pass, GPTZero. Reported methods, numerical results, citations and qualifications were checked against the source; an inaccurate description of the noise-reassignment guard was corrected against the implementation. Detector scores were used as editing feedback; they do not verify authorship or scientific validity. Some blocks remained above the editing target after repeated revisions.

Claude was used as follows:

- To write code for Python, for performing data cleaning/preprocessing, imputation, clustering and visualization, and debugging.
- To assist with drafting: editing draft versions to improve academic tone/clarity and rephrase initial drafts.
- To develop dashboard: assisting in developing the Streamlit interactive dashboard (CogDash).
- To format LaTex document: formatting of thesis document (LaTex), tables, figures, compiling issues.

I take responsibility for designing/researching hypothesis creation, methodology development, data analysis and interpretation/discussion. I reviewed/edit all generated AI-texts myself and verified referenced materials with source documents.

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Chapter 1

Introduction

1.1 Background and Problem Statement

Acquired Brain Injuries (ABIs) consist of traumatic brain injuries (TBIs), stroke, encephalitis and hypoxic-ischemic injury. The clinical implications of ABIs are large. TBIs account for millions of emergency department visits/hospital admissions every year [Maas et al., 2017; Brazinova et al., 2021] ; ABI can also cause long-term disabilities for adults of working age, with physical, behavioral/emotional and cognitive changes [Corrigan et al., 2010]. The areas that are typically impacted by ABI include attention, memory, language and executive functioning but vary significantly from one patient to another.

The variation makes it difficult to develop individualized treatment/intervention plans based upon cognitive assessments of ABI patients. Post-ABI follow-up evaluations can identify patients who have sustained injuries of the same severity and yet demonstrate vastly different cognitive patterns and abilities. Commonly used criteria (diagnostic category or severity banding) grouping together different cognitive profiles will obscure clinically meaningful differences between individuals with identical diagnostic/severity categories [Saatman et al., 2008].

This research aims to investigate the above-stated concerns empirically. As opposed to applying previously established clinical taxonomic criteria to the dataset used for this research, I apply an unsupervised clustering method to group patients by their performances as measured on multiple neuropsychological tests, and then examine the extent to which the groups (subgroups) obtained from such application are clinically relevant. Another common obstacle to conducting analyses with clinically-derived data involves missing values. Clinicians often leave out components of neuropsychological testing batteries when administering them due to time limitations, patient fatigue, clinician judgment, or patient-related issues which may impede the completion of all test administrations. Non-random omissions of some test items are therefore anticipated.

Missing value imputations allow for retaining incomplete assessments; however, each procedure has its own assumptions about missing values. There is minimal documentation regarding how those assumptions affect subsequent cluster analysis,

thus the relationship of those assumptions to cluster analysis is investigated directly within this dissertation.

The process of phenotyping represents the identification of cognitive subgroups through a data-driven process. The question remains unanswered as to whether these subgroups constitute qualitative differences in cognitive profiles or represent continuum levels of general severity.

1.2 Research Questions

The purpose of this study was to answer the following four research questions:

Research Question RQ1: Is it possible to isolate discrete cognitive phenotypes through the use of an unsupervised clustering method applied to the imputed dataset on participants with a history of an acquired brain injury?

Research Question RQ2: How much does the choice of an imputation methodology impact upon the identification of cognitive phenotypes?

Research Question RQ3: Are there associations between patient cluster assignment and clinical unit assignment? Such an association would suggest that cognitive tiers differ in their distribution among the populations served by different units.

Research Question RQ4: Can either (a) increased stability in grouping or (b) improved interpretation occur if the clustering utilizes composite scores representing domain areas as opposed to using individual tests in the clustering?

The objective ultimately is to create a reproducible phenotyping procedure based on routine clinical data collection. Missingness will be modeled as part of the rehabilitation process rather than treated as a nuisance only after data collection.

1.3 Hypotheses

I test four hypotheses in this dissertation.

Hypothesis H1: My expectation using the UMAP/HDBSCAN pipeline is that there will be at least two cognitively stable clusters for most imputation methods. Also, I expect at most 30% of all data points to be labeled as noise, and the average silhouette score will be at least 0.40.

Hypothesis H2: I used the "fixed reference" from Chapter 3 to test if group assignments were consistent across different imputation methods. Specifically, I expected the median pairwise ARI between imputations to exceed 0.50. As stated previously, I would view this as evidence that the assignment is robust to these specific imputed data sets, but it would not rule out artifacts shared by the representation or establish clinical phenotypes.

Hypothesis H3: Chi-Square tests assessing association between cognition cluster membership and Clinical Unit population should show statistically significant results. I expect the value of Cramér's V to be greater than 0.10. These results would demonstrate differences among the populations served by clinical units.

Hypothesis H4: By aggregating domain specific features, I expect the same level of separation among groups, but lower levels of redundancy. Metrics that I will use to determine if this is accurate include silhouette values, Variance Inflation Factor (VIF), and Condition Number (CN).

An exploratory patient-level extension followed the historical H1-H4 analyses. It independently refits clustering across imputations and patient subsamples, compares feature scaling and incomplete-data mixture models, and models longitudinal cognitive scores. Methods and results are in Sections 3.8 and 4.12. It does not retrospectively redefine the original hypotheses.

Chapter 2

Literature Review

2.1 Acquired Brain Injury and Cognitive Assessment

Acquired brain injury (ABI) refers to brain damage sustained after birth that is not due to a congenital or degenerative condition [UK Department of Health and Social Care, 2022]. It includes traumatic and non-traumatic injuries such as traumatic brain injury (TBI), stroke, encephalitis and hypoxia/ischemia. Although there are many different etiologies of ABI, their commonality lies within their shared rehabilitative challenges and outcomes. Cognitive deficits post-injury will frequently influence long term functional status. One way that clinicians assess cognitive deficits that result from these types of injuries is through administering standard, normed neuropsychological tests. Most standard test batteries are designed to assess individuals' abilities across multiple domains of cognition. This includes assessing an individual's ability to visually/spatially process information, process language, engage in tasks requiring executive functions, learn and retain new information, and finally, how quickly an individual can process information. Each domain assessed has practical implications. Examples include, but are not limited to, attention deficit impeding an individual's participation in therapeutic interventions, memory impairments limiting an individual's ability to develop effective compensatory strategies for daily living activities, limitations in executive and language skills contributing to issues with planning, self-advocacy and communicating needs, and visuospatial impairments impacting an individual's ability to independently navigate their environment. Thus, a cognitive profile generated through comprehensive administration of neuropsychological testing can provide valuable information for planning rehabilitation programs, assist in establishing medicolegal issues related to the injury, and inform return-to-work decisions.

People who suffer from moderate to severe traumatic brain injuries (TBI) can have lasting cognitive problems, and there is no one way for people to recover from their injuries. Long-term studies have examined functioning for up to ten years after injury [Ponsford et al., 2014]. In their review, Rabinowitz and Levin [2014] describe that the cognitive outcome depends upon a number of factors related to the injury itself as well as the patient's history, including the patient's age at time of injury and pre-injury neuropsychiatric status. Injury severity alone therefore cannot fully

describe an individual's cognitive profile.

Neuropsychologists will typically use modified versions of standard test batteries based upon such variables as: what specific questions were posed in the referral; the patient's tolerance for testing; and clinical judgment. Each of these decisions introduces additional variability/structured missingness into the resultant data sets. Therefore, multivariate analyses need to take this structured missingness into account rather than simply viewing the missing values as merely a minor methodological concern.

2.2 Missing Data in Clinical Research

Although missing data occur frequently in clinical research [Gravesteijn et al., 2021], three mechanisms are distinguished [Rubin, 1976]: missing completely at random (MCAR), missing at random (MAR) and missing not at random (MNAR). Missing completely at random (MCAR) indicates that missingness is unrelated to all values (both seen and unseen). Missing at random (MAR) implies that missingness may depend upon the observed values, but once those values are accounted for, then there will be no dependence on the missing values. Missing not at random (MNAR) suggests that missingness remains dependent upon the unseen values even when the observed values are considered.

Each type of missingness affects what statistical methodologies should be employed to produce reliable estimates. Under MCAR, listwise deletion is unbiased, but reduces the sample size and usually the efficiency of the study. When MAR exists, both multiple imputation and maximum likelihood estimation allow for valid inference under an appropriately constructed model. MNAR models require sensitivity analysis because observed data alone cannot establish assumptions about unobserved values [Rubin, 1987].

This understanding is especially important within clinical neuropsychology. Consider someone with severe attentional impairment who lacks the energy to complete an executive-function task, or a disoriented patient who is not given a memory test. In both examples, the missing value may be related to the value that would exist if the full test had been administered. Therefore, MAR is merely assumed as opposed to being definitively proven. MNAR patterns can coexist with missingness caused by assessment protocols. Although observed data alone cannot identify the missingness mechanism, auxiliary variables can make MAR more plausible by incorporating clinically relevant information [Little and Rubin, 2019].

The amount of missing data alone cannot establish the risk of bias. Instead, it is the reason(s) for which the data are missing along with observed associations among other measures that influence the quality of imputed values. To provide sufficient evidence for conditional estimation, variables should be selected to include in the analysis such that there exists enough observed data on each variable. Subsequent

cluster analyses should examine how imputation decisions affect the structure recovered.

2.3 Imputation Methods

The ten methods presented in this study are ordered by increasing complexity. Let the data matrix $X \in \mathbb{R}^{n \times p}$ contain n assessments and p variables. For assessment i, x_{ij} denotes the value of variable j. Let $\Omega = \{(i,j) : x_{ij} \text{ is observed}\}$. Then $\Omega_j = \{i : (i,j) \in \Omega\}$ represents the observed rows for the j-th variable. Define the operator P_Ω as follows: for any entry x_{ij} of the matrix X, $P_\Omega(x_{ij}) = x_{ij}$, if $(i,j) \in \Omega$; otherwise, $P_\Omega(x_{ij}) = 0$. The task is to estimate the entries outside Ω, with estimates denoted by $\hat{x}_{ij}$.

Mean Imputation Method.

Mean imputation calculates the average of the observed values of a variable and assigns that mean to each missing value of the same variable.

$$\hat{x}_{ij} = \bar{x}_j = \frac{1}{|\Omega_j|} \sum_{i \in \Omega_j} x_{ij}. \tag{2.1}$$

Mean imputation is extremely rapid and is non-iterative. However, Mean imputation reduces variance and covariances among variables, alters marginal distribution of variables and decreases estimated standard error of parameters. Therefore, Mean imputation was employed exclusively as a reference point or "baseline" for purposes of this investigation.

K-Nearest Neighbors (KNN).

For each missing value, KNN imputation finds k nearest "donor cases" based upon Euclidean distance to define the degree of similarity. Then, KNN imputes a weighted average of values from these donor-cases into the missing values. The weights are inversely proportional to the distances, as follows:

$$\hat{x}_{ij} = \frac{\sum_{k \in \mathcal{N}_i} w_{ik}\, x_{kj}}{\sum_{k \in \mathcal{N}_i} w_{ik}}, \qquad w_{ik} = \frac{1}{d(i,k)}, \tag{2.2}$$

where $\mathcal{N}_i$ is the set of k-nearest neighbors and $d(i,k)$ is the distance between observations in the jointly observed features. KNN is non-parametric and can capture local data structure more effectively than mean imputation. When using KNN as an imputation method you need to be careful when you are choosing your k. If you choose too low of a k, you may be fitting the noise instead of the signal. If you select too large of a k, you may "smooth" away important patterns in your data. Larger datasets require longer computation times, which is a consideration for larger datasets.

Multiple Imputation by Chained Equations (MICE).

MICE (Multiple Imputation by Chained Equations) is used for imputation of each variable in relation to all other variables in the dataset. However, each variable is iteratively imputed, over multiple iterations [Buuren and Groothuis-Oudshoorn, 2011]. Each iteration t draws new values for each variable from its conditional distribution:

$$x_{ij}^{(t)} \sim p\left(x_{ij} \,\middle|\, \mathbf{x}_{i,-j}^{(t)}; \theta_j\right), \qquad j = 1, \ldots, p, \tag{2.3}$$

Depending on how you want to represent your variable(s), you may define different types of conditional models; i.e., linear models for continuous variables or logistic models for binary variables. The advantage of MICE is that it can maintain relationships between multiple variables and generate multiple imputations to allow you to measure uncertainty. Some disadvantages include defining a model for every variable and possible convergence problems.

Predictive Mean Matching (PMM).

PMM (Predictive Mean Matching) is commonly used in conjunction with MICE (i.e., as a semi-parametric imputation method). First, a regression model is created to predict values. Then, a missing value is replaced by a random selection of an observed donor value with a comparable model prediction:

$$\hat{x}_{ij} \sim \text{Unif}\{x_{kj} : k \in \mathcal{D}_{ij}\}, \qquad \mathcal{D}_{ij} = \{\, d \text{ observed } k \text{ minimising } |\hat{x}_{kj} - \hat{x}_{ij}| \,\}. \tag{2.4}$$

Predictive mean matching selects observed donor values, keeping imputations within the observed range. This avoids impossible values, such as a negative score on a test with a floor. A limited donor pool can lead to repeated use of the same values.

MissForest.

MissForest utilizes iterative random forests for its missing value imputation algorithm [Stekhoven and Bühlmann, 2012]. For the first step, for every variable j, a forest f_j is created based on all instances of j where j is known. All other variables are treated as predictor variables. Next, a missing value is replaced by using the newly constructed forest f_j. The process repeats until changes are small.

$$\hat{x}_{ij} = f_j(\mathbf{x}_{i,-j}) \quad \text{until} \quad \sum_{(i,j)\notin\Omega} \left(x_{ij}^{(t)} - x_{ij}^{(t-1)}\right)^2 \text{ stops decreasing.} \tag{2.5}$$

Since MissForest does not assume parametric distributions, it allows the discovery of complex relationships between variables and can also be applied to either continuous or categorical variables. Generally, MissForest often performs well on benchmark datasets. A significant drawback of MissForest is that it can be quite

time consuming and its computational efficiency decreases rapidly with increasing amounts of data or number of trees.

Expectation-Maximization (EM).

Another iterative method is expectation-maximization (EM) [Dempster et al., 1977]. EM works by alternating between two different types of steps: An E-step which estimates the expected sufficient statistic, and an M-step which maximizes the likelihood function. Under a multivariate normal model, the iterations estimate model parameters and missing blocks can be filled using their conditional means:

$$\hat{\mathbf{x}}_{\text{mis}} = \boldsymbol{\mu}_{\text{mis}} + \Sigma_{\text{mis,obs}} \Sigma_{\text{obs,obs}}^{-1} (\mathbf{x}_{\text{obs}} - \boldsymbol{\mu}_{\text{obs}}) \,, \tag{2.6}$$

where $\boldsymbol{\mu}$ and Σ are recomputed after every iteration. We should note that assuming a multivariate normal distribution may be unrealistic when there are many highly skewed or bounded variables.

SoftImpute.

SoftImpute uses Singular Value Decomposition (SVD) with soft-thresholding to create a low-rank approximation of the original input matrix [Mazumder et al., 2010]. Specifically SoftImpute solves:

$$\min_{Z} \ \tfrac{1}{2} \big\| P_{\Omega}(\mathbf{X} - Z) \big\|_F^2 + \lambda \|Z\|_*, \tag{2.7}$$

where $S_\lambda(Z) = U \operatorname{diag}\big((d_i - \lambda)_+\big) V^\top$. λ determines how much to shrink the singular values and therefore the rank of the estimated matrix. While SoftImpute can work efficiently on very large sparse matrices, it does not necessarily capture structure in the data that cannot be captured through a low rank estimate.

Non-Negative Matrix Factorization (NMF).

Like SoftImpute, Non-Negative Matrix Factorization (NNMF), or NMF, also uses lower dimensional representations of the data. NMF creates two non-negative matrices [Lee and Seung, 1999] that together represent the original input matrix:

$$\min_{W \geq 0, H \geq 0} \big\| P_{\Omega}(\mathbf{X} - WH) \big\|_F^2, \qquad W \in \mathbb{R}_{\geq 0}^{n \times r}, \ H \in \mathbb{R}_{\geq 0}^{r \times p}. \tag{2.8}$$

NMF preserves the constraint when test scores are non-negative. The factors produced by NNMF provide an additive way of representing the score matrix. Sometimes, the factors of the score matrix can have meaningful interpretations. However, like all other methods described here, finding a good number of factors (i.e. determining a suitable r) is critical for producing useful results.

Denoising Autoencoder (DAE).

Within the MIDA paradigm, Denoising AutoEncoders (DAE's) learn a Neural Network to take in corrupt versions of the data and output reconstructions of the

data [Gondara and Wang, 2018]. For an encoder f_θ, decoder g_θ and corrupted version of the data $\tilde{\mathbf{x}}$, they try to find the minimum distance between their reconstruction of the original input and the original input:

$$\mathcal{L}_{\text{DAE}}(\theta) = \mathbb{E}\left\|\mathbf{x} - g_\theta(f_\theta(\tilde{\mathbf{x}}))\right\|^2. \tag{2.9}$$

The resulting imputed values are determined by averaging the values generated by multiple neural-network models, each initialized with a different random seed. Denoising autoencoders may outperform traditional imputation techniques when the relationships among variables are nonlinear.

Variational Autoencoder (VAE).

For example, VAEs use probabilistic generative models to define probability distributions in latent variable space. In so doing, observations are represented as distributions over latent variables [Kingma and Welling, 2014; McCoy et al., 2018]. The evidence lower bound (ELBO) can be maximized by:

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \beta\, D_{\text{KL}}\left(q_\phi(z \mid x) \,\|\, p(z)\right), \tag{2.10}$$

using a standard normal prior $p(z) = \mathcal{N}(\mathbf{0}, \mathbf{I})$. In related work, [Mattei and Frellsen, 2019] used importance-weighted bounds for missing data, whereas [Yoon et al., 2018] proposed a GAN-based method. Using VAEs provides an inherent way of producing imputed values based on samples from the latent distribution, followed by decoding each sample. If trained properly this also allows for a representation of the uncertainty of each individual value.

Deep learning techniques have advantages and disadvantages. They may take more computation time than simpler methods, but can learn nonlinear relationships without a manually specified simple functional form for each variable. Many architectures also contain stochastic elements: dropout in DAEs and latent sampling in VAEs are examples. These elements can be used to generate multiple imputations. However, deep learning models can overfit and their performance often depends heavily on architectural choices. This dissertation examines how these methods function on the small to medium-sized, relatively low-dimensional datasets typical of clinical neuropsychology [Prakash et al., 2024].

[Afkanpour et al., 2024] divided imputation methods for clinical research into four categories. I add a fifth for the deep learning methods used here. Categories one through three are comprised of the well-known methodologies which include: Category One: Traditional Methods (mean imputation, EM, MICE); Category Two: Machine Learning Methods (KNN, MissForest); Category Three: Hybrid Methods (PMM). Matrix Completion represents Category Four (SoftImpute and NMF). Category Five was created specifically for Deep Learning Methodologies utilized in our study (DAEs and VAEs). Utilizing these five categories will allow us to determine

whether the methodologies in each category provide consistent recovery of cognitive structures when utilizing each methodology with varying assumptions about estimating missing data.

2.4 Clustering in Neuropsychology

Clustering is used as a statistical technique in neuropsychological research. Clinical judgments about diagnosis or disability after stroke, traumatic brain injury (TBI), or in mild cognitive impairment (MCI) often rely on categorical assessments. Across many different types of clinical populations (e.g. TBI; MCI), cluster analysis has been used. Many of the resultant analyses have indicated that the statistically derived groups of performance within the realm of cognition do not consistently match the clinical diagnosis or severity category that are given to those same populations. Therefore, from an empirical perspective, it appears that when clinicians utilize classification systems to describe cognitive variability among individuals post-injury, they do not fully capture the full range of cognitive variability present.

A literature review of data-driven cognitive phenotyping after brain injury was conducted by [García-Rudolph et al., 2021]. Trends were consistent in their reviews. First, many of the studies found two to four statistically defined cognitive clusters. Second, many of the studies found clusters based upon whether certain cognitive domains were more functional than others relative to an individual's overall level of functioning. Third, when possible and relevant, researchers found that group membership was significantly related to clinically important outcomes (i.e. return to work).

Many aspects of research on data-driven cognitive phenotyping following brain injury are still underdeveloped. Many prior studies have been based on samples that ranged from approximately 100 to less than 200 subjects. The common practice among researchers was to exclude missing values by using complete cases. When missingness is MCAR (completely at random) this approach may be suitable for determining the characteristics of the remaining sample; however, when there is some degree of information within the missingness, then the use of complete cases can bias the resulting sample. Very few studies have investigated how sensitive clustering algorithms are to various parameter choices.

Therefore, this dissertation employed four distinct approaches to reduce the impact of these limitations. In Chapter 3 I used a much larger clinical database; compared ten imputation strategies to each other and to complete case analyses; examined sensitivity to hyperparameters and to several missingness thresholds; and tested HDBSCAN as an alternative method to k-means. Unlike k-means, HDBSCAN does not require the number of clusters k to be specified before application. Instead HDBSCAN identifies clusters where the density of data points exceeds certain threshold(s); data points which do not fit into a cluster are identified as "noise" or outliers.

2.5 UMAP and HDBSCAN

The clustering process is composed of two key components: the UMAP for reducing the dimensionality of the input dataset and identifying clusters using HDBSCAN.

2.5.1 UMAP

The first part (UMAP) applies a type of non-linear manifold learning. It seeks to reduce high-dimensional datasets to lower-dimensional representations while retaining as much of their original structure as possible. Because PCA and other forms of linear reductions cannot capture non-linear relationships present in the original data, UMAP can. As such, there are trade-offs involved with preserving the overall topological features of the data (i.e., globally) versus the local details, so we use three parameters to find a suitable balance between global and local structure:

- `n_neighbors`: This determines the number of neighboring points in the original data that should be used in the construction of each point's neighborhood graph. The larger this value is, the broader your view will be of the global properties of the data. Conversely, if you choose smaller values, you will have better insight into local relationships.
- `min_dist`: This governs how closely points can be placed together in the reduced space. If you increase this value, it results in spreading points farther apart from one another; conversely, setting it to a smaller value allows for denser grouping of related points.
- `n_components`: This is the target number of dimensions in the reduced space.

UMAP creates a weighted graph of neighboring objects. Each connection has a directed membership probability $p_{j|i}$ linking an object x_i to one of its neighbors x_j, as defined in Equation 2.11. Two parameters enter this calculation: ρ_i, the distance from x_i to its closest neighbor, and σ_i, a calibration factor chosen so that $\sum_j p_{j|i} = \log_2(k)$, with k specified by `n_neighbors`. The directed probabilities are made symmetric using Equation 2.12. In Equation 2.13, q_{ij} is the low-dimensional membership probability, with a and b determined by `min_dist`. Lastly, UMAP minimizes the cross-entropy given in Equation 2.14 between the high-dimensional and low-dimensional memberships.

$$p_{j|i} = \exp\left(-\frac{\max\{0,\, d(x_i, x_j) - \rho_i\}}{\sigma_i}\right), \tag{2.11}$$

$$p_{ij} = p_{j|i} + p_{i|j} - p_{j|i}p_{i|j}. \tag{2.12}$$

$$q_{ij} = \left(1 + a\,\|y_i - y_j\|_2^{2b}\right)^{-1}, \tag{2.13}$$

$$C = \sum_{i \neq j} \left[p_{ij} \log \frac{p_{ij}}{q_{ij}} + (1 - p_{ij}) \log \frac{1 - p_{ij}}{1 - q_{ij}} \right], \tag{2.14}$$

2.5.2 HDBSCAN

HDBSCAN represents data through a hierarchy of clusters at varying levels of density and identifies groupings that persist across that hierarchy. With respect to the recovery of variable-density and nested clusters, DBSCAN suffers in comparison because it uses a constant radius. HDBSCAN's explicit density hierarchy addresses this limitation. It selects stable clusters without requiring their number in advance and assigns the remaining observations to noise. Three key parameters govern its operation:

- `min_samples` - Density estimation parameter.
- `min_cluster_size` - Minimum size of a valid cluster.
- `cluster_selection_epsilon` - Distance threshold used to merge nearby clusters.

To connect observations, HDBSCAN uses mutual reachability (Equation 2.15), which adjusts distances to reflect local density. A minimum spanning tree over those distances provides the basis for the hierarchy. Successive levels form groups of observations, and cluster persistence is assessed through stability (Equation 2.16). For a given cluster C, $\lambda_{\text{birth}}(C)$ is the density level at which C first appeared; $\lambda_{\max}(x, C)$ is the highest density level at which x belonged to C. Observations outside the selected persistent clusters are assigned to noise.

$$d_{\text{mreach}}(a, b) = \max\{\text{core}_k(a),\ \text{core}_k(b),\ d(a, b)\}. \tag{2.15}$$

$$S(C) = \sum_{x \in C} \left(\lambda_{\max}(x, C) - \lambda_{\text{birth}}(C)\right), \qquad \lambda = \frac{1}{d_{\text{mreach}}}, \tag{2.16}$$

The ability to extract structure from high-dimensional datasets has led to applications of UMAP and HDBSCAN in computational biology and related fields.

2.6 Cluster Validation

The current research uses three different techniques in order to evaluate if the cluster structures produced by these methods are appropriate since they have no gold standard for reference. Internal validation methods, external consistency measures, as well as sensitivity analysis are used to indirectly assess the validity of this research.

2.6.1 Internal Validation

Internal validation involves several aspects of cluster analysis. In this research we have used silhouette measures [Rousseeuw, 1987] to provide comparison of how well points in a cluster are cohesive (i.e., similar), with respect to how separated they are from points in their nearest neighbor cluster(s). A silhouette value will be a number in [-1,+1]. Silhouette scores greater than 0 indicate that a point is closer, on average, to other points assigned to the same cluster than to any other cluster. Scores close to 0 indicate little difference among those alternatives; while negative scores indicate that the point is actually closer to another cluster. These values do not determine whether an assignment is clinically correct. Each silhouette measure is calculated as per Equation 2.17, and is composed of two components: $a(i)$; the mean distance between each point and every other point in the same cluster; and $b(i)$; the mean distance from the point to every member of the nearest alternative cluster. The overall average silhouette for a set of clusters can be found by calculating the average of all $s(i)$ values and will serve as our primary method of internally evaluating the quality of the clusters generated. Noise percentage is also reported. It is the proportion of observations left unassigned by density clustering and therefore describes its coverage.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i),\, b(i)\}}. \tag{2.17}$$

2.6.2 External Validation

Comparing cluster models for the same observations (e.g. cluster assignments generated using different imputation techniques), can be achieved by evaluating the degree of concordance among the resultant cluster models. An effective way to perform this evaluation is through the application of the adjusted Rand index (ARI) [Hubert and Arabie, 1985]. This metric evaluates the degree of concordance between the pair-wise cluster assignments to evaluate if the concordance exceeds chance.

To utilize the formula shown in Equation 2.18, we must first identify two clusterings of n objects denoted as $U = \{U_i\}$ and $V = \{V_j\}$. We will also need to identify contingency counts denoted as $n_{ij} = |U_i \cap V_j|$, where i and j index the clusters in U and V. The marginal totals denoted as $a_i = \sum_j n_{ij}$ represent the number of elements in U_i. The marginal totals denoted as $b_j = \sum_i n_{ij}$ represent the number of elements in V_j.

$$\text{ARI} = \frac{\sum_{ij}\binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]\Big/\binom{n}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]\Big/\binom{n}{2}}. \tag{2.18}$$

When ARI is maximally positive (+1) there exists a complete concordance between both cluster models. Conversely, ARI values which equal 0 indicate that the

concordance observed between both cluster models equates to a chance baseline. Finally, negative values of ARI will indicate less than expected concordance between both cluster models [Robert et al., 2020].

In addition to the usage of ARI I also make use of normalized mutual information (NMI). NMI quantifies how much one cluster model provides information about another clustering, while providing normalization with respect to the entropy of each cluster model label.

The results of comparing the cluster assignment solutions through mutual information (i.e., $I(U;V)$) and Shannon entropy (i.e., $H(\cdot)$) [Shannon, 1948] as expressed in Equation 2.19 can range from 1 when there is perfect agreement among cluster assignments to 0 when they are found to be statistically independent.

$$\text{NMI}(U,V) = \frac{2\,I(U;V)}{H(U)+H(V)}, \tag{2.19}$$

To measure the degree that clusters in one solution are either similar or dissimilar to those in another solution, I utilize the Jaccard Index. As indicated in Equation 2.20, where A and B represent the sets of observations that comprise the clusters being evaluated for similarity, the Jaccard Index represents the size of the intersection of the sets divided by the size of the union of the sets, with values in [0, 1]. Therefore, larger values of the index would indicate that the cluster memberships are more similar.

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}. \tag{2.20}$$

In addition to evaluating the similarity/dissimilarity between clusters across two solutions, I utilize clusterwise stability matching [Hennig, 2007], to evaluate whether cluster assignments remain stable after repeated sampling. In each of R resampling iterations, I pair an original cluster C_k with the resampled cluster having the greatest Jaccard overlap with it. I then average these maximum overlaps across the iterations. This average value is referred to as the "stability score" for C_k and is expressed as in Equation 2.21. A per-cluster score > 0.7 is used here as the stability criterion.

$$\hat{S}_k = \frac{1}{R}\sum_{r=1}^{R} \max_{k'} J\left(C_k, C_{k'}^{(r)}\right). \tag{2.21}$$

Taken together, NMI and ARI enable a quantitative comparison of similarities and differences between two cluster assignments. Although both metrics enable an evaluation of agreement between two cluster assignments they are metrics that measure agreement and should not be used as statistical tests of significance.

Chapter 3

Methodology

3.1 Data Description

There were 22,075 raw assessments for 8,739 individuals within a clinical neuropsychology database that is managed by a rehabilitation center. Some of these individuals had assessments completed at different times. Within each assessment, there are 31 variables collected; they include demographics, administrative information, information related to their injury, and test results across six cognitive domains: orientation, attention, visuoperception (visuospatial), language, memory, and executive function. Following Tier-1 filtering as discussed in Section 3.2, the number of assessments remaining after filtering was 17,406, with 7,285 unique individuals included in the final analyzed sample of data.

In terms of the historical aspect of this study, the single assessment level represented the unit of observation. The patient-level extension in Section 3.8 uses each patient's earliest eligible assessment as the index assessment and retains the same discovery/validation patient split for trajectory modeling. For the historical analyses assessing clustering, imputation comparison, and assessment level bootstrap methods, each follow-up assessment was treated separately (as separate observations). The assessments were not aggregated into units representing the 7,285 patients. Although treating each assessment individually preserves those observed changes, it does not acknowledge the dependency that exists between consecutive assessments of the same individual. Therefore, the resulting stability estimates may appear to be more precise than if estimates were derived from resampling grouped by patient. The primary exception to these historical analyses was the multinomial sensitivity analysis located in Section 4.7, which involved grouping both the training/test sets and bootstrap samples by patient.

The data come from an actual clinical environment. Therefore, in assessing which measures to administer, the psychologist making those decisions made use of their own professional judgment. They selected measures that would address questions related to the clinical issue at hand; and they did so taking into account whether or not the patient was capable of participating. The data then exhibit systematic patterns of missing data. While it is true that there are many measures that appear on most evaluations; many others will only appear when a particular clinical

concern arises. In consequence, the rate of missing data across the various measures available for evaluation is indicative of the clinicians' decision-making process regarding how to assess the patient's clinical condition. Rates of missingness range from approximately 5% for screening tools to greater than 90% for specialty assessments.

Variables in the dataset also differ significantly in terms of the percent of missing values each variable contains. Variables additionally differ from one another in terms of measurement scale (i.e., standardized z-scores/scaled scores/timed raw scores). Scales were aligned during domain aggregation (Section 3.4). Table 4.1 displays the distribution of missing data among the 14 candidate variables. It can be seen that 53.1% of the 17,406 assessments (n = 9,245) contained complete data for all 14 variables. Thus, if complete-case analysis were performed, it would result in exclusion of 46.9% (8,161 assessments) of the sample size. The imputation approach (Section 3.3) is designed to allow retention of these cases and their associated clinical observations.

All data have been deidentified using standard procedures to protect patient confidentiality.

3.2 Tier 1 Filtering

The selection criteria for variables were based upon a threshold of 50% missing values so that the results of the analysis would be limited to the 14 cognitive tests listed below. Each assessment was required to have at least one observed score among these 14 retained tests. The 50% missingness threshold therefore pertained to variables, but the eligibility requirements were established by the minimum number of observed scores in an assessment.

This resulted in a total of 17,406 assessments with data from 7,285 patients. There were 16,553 assessments which contained at least 7 observed test scores among the 14 tests and there were 853 assessments containing less than 7 observed test scores. The sensitivity to the historic thresholds used to select variables is discussed in Section 4.10. The patient-level sensitivity analysis which requires at least 7 observed test scores is included separately in Section 4.12. Applying a stricter row rule changes the cohort and cannot be described as the rule that created the original historical data matrix.

3.3 Imputation Methods

The rest of this subsection describes ten methods of imputation. However, not all of these are multiple imputation methods. For the clustering evaluation, I gave the silhouette value and number of clusters k that provided the best results using UMAP + HDBSCAN with domain-level feature sets. Here, I give the silhouette value based only upon the high-density points that were identified and grouped by HDBSCAN

TABLE 3.1: Eligible neuropsychological variables after Tier 1 filtering, organised by cognitive domain.

Cognitive Domain	Variables	Count
Orientation	OPERSONA, OESPAI, OTEMPS	3
Attention	ASPAN, ATMTA	2
Visuoperception	VPIMATGES	1
Language	LREPETICIOTB, LDENOMINACIOTB, LCOMPRENSIOTB	3
Memory	MDIGITS, MRAVLT075, MRAVLT015, MRAVLT015R	4
Executive Function	FEPMR	1
Total		**14**

before noise points were assigned to their nearest neighbors. This differs from evaluating all assigned cases, as in Section 4.5: MICE gave a silhouette of 0.598 before noise reassignment and 0.526 across all assigned cases.

For the two deep learning methods used to generate estimates of missing values, there were also other options made available regarding their training. Therefore, technical details of these models will be provided below. The Denoising AutoEncoder (DAE) was trained on corrupted input, where observed values were masked. In order to create the corrupted inputs, masks were randomly selected according to the following distribution:

$$\mathbf{m}_i \sim \begin{cases} P_{\text{miss}} & \text{with probability 0.5,} \\ \text{Bernoulli}(1-\rho)^{\otimes p} & \text{with probability 0.5,} \end{cases} \tag{3.1}$$

Here, P_{miss} denotes the empirical distribution of missingness patterns and ρ is the uniform corruption rate. This allowed the DAE to train on random corruption as well as missingness patterns like those in the clinical data.

The VAE used a cyclical annealing schedule for the β term in its evidence lower bound:

$$\beta(t) = \beta_{\max} \frac{t \bmod T}{T}, \qquad T = 40,\ \beta_{\max} = 0.1, \tag{3.2}$$

Deep-learning imputation methods were assessed by removing a subset H of the observed entries and comparing the original values to those which had been estimated through imputation. Root mean square error (RMSE) and mean absolute error (MAE) were calculated using the following equations:

$$\text{RMSE} = \sqrt{\frac{1}{|H|} \sum_{(i,j)\in H} \left(\hat{x}_{ij} - x_{ij}\right)^2}, \qquad \text{MAE} = \frac{1}{|H|} \sum_{(i,j)\in H} \left|\hat{x}_{ij} - x_{ij}\right|. \tag{3.3}$$

TABLE 3.2: A summary of the ten imputation methods broken down by the six-category taxonomy. The clustering performance columns report the optimal UMAP+HDBSCAN solution on domain-level features, using the high-density HDBSCAN core before k-nearest-neighbour noise assignment.

Category	**Method**	**Year**	k	**Sil.**	**Advantages**	**Limitations**
Conventional	Mean	—	4	0.604	Simplest; fast; universal baseline	Collapses variance & covariance
Statistical	EM	1977	4	0.587	Principled MLE; preserves covariance	Assumes multivariate normality
	MICE	2011	3	0.598	Flexible; preserves distribution	Requires model per variable
Machine	KNN	2001	4	0.576	Non-parametric; captures local structure	Sensitive to k; slow for large n
Learning	MissForest	2012	3	0.606	Non-linear; handles mixed types	Computationally expensive
Hybrid	PMM	—	4	0.498	Preserves marginal distribution	Requires adequate donor pool
Matrix	SoftImpute	2010	4	0.571	Efficient for sparse matrices	Assumes low-rank structure
Completion	NMF	1999	3	0.546	Non-negativity natural for scores	Sensitive to rank choice
Deep	DAE	2018	5	0.510	Non-linear; multiple imputations	Requires hyperparameter tuning
Learning	VAE	2014	4	0.603	Probabilistic; principled uncertainty	Complex training; β-tuning

References: Little and Rubin [Little and Rubin, 2019]; PMM and MICE [Buuren and Groothuis-Oudshoorn, 2011]; EM [Dempster et al., 1977]; KNN [Troyanskaya et al., 2001]; MissForest [Stekhoven and Bühlmann, 2012]; SoftImpute [Mazumder et al., 2010]; NMF [Lee and Seung, 1999]; DAE [Gondara and Wang, 2018]; VAE [Kingma and Welling, 2014; McCoy et al., 2018].

3.4 Calculating Scores for Each Cognitive Domain

A collection of cognitive-domain features was additionally generated by aggregating of the variables in each area. The same 14 individual neuropsychological variables were retained for comparison. Several of the measures used different scales (i.e., standardized scores, scaled scores). Without scale alignment, those differences could cause problems during aggregation. To reduce problems from extreme imputed values associated with outliers or sentinel codes, three processing steps were completed on each variable before aggregation:

1. Winsorizing each variable at the 1st and 99th percentile.
2. Inverting signs for time-based assessments to make higher scored values representative of superior performance.
3. Standardizing each variable to have a mean of 0 and variance of 1.

The overall scores for the domains were determined by calculating an administration-frequency-weighted mean of the variables that represent the respective domain. Each variable j received a weight $w_j = 1 - r_j$, with r_j representing its missingness rate; weights were then normalized within each domain. Thus, variables that are measured more often will have a higher weight and therefore have a stronger impact when computing the domain score. In addition, the less frequent variables are reduced in their contribution to the domain score due to reliance on imputation of missing data. These weights do not measure psychometric reliability: they estimate neither internal consistency nor test-retest reliability or measurement error. Aggregating fourteen variables into six domain scores gives distance-based methods fewer dimensions to operate over. Clinicians who evaluate patient cognitive function based on a composite of multiple tests may find evaluating cognitive domains (e.g., attention, executive functions) easier to interpret than assessing individual test results. However, there is a trade-off in that two individuals receiving equivalent average memory scores may still demonstrate differences in immediate recall vs delayed recall. Hypothesis H4 addresses whether the reduction in redundancy and improvements in interpretation provide sufficient justification for accepting a loss of detail.

3.5 UMAP + HDBSCAN Pipeline

Figure 3.1 provides a comprehensive depiction of all steps taken from imputation through final cluster designation. In order to decrease sensitivity of the UMAP (and subsequently the HDBSCAN) to the extremes in the neuropsychological assessment scores I perform robust scaling using the median and interquartile range [Bellio et al., 2020; Pinsky and Klawansky, 2023] before performing UMAP and HDBSCAN. When floor effects are present in an assessment, severely impaired individuals will

tend to accumulate at the lower end of the scales used in that assessment [Pinsky and Klawansky, 2023]; ceiling effects occur when those who have no impairment accumulate at the upper end of the scales for that same assessment [Pinsky and Klawansky, 2023]. Robust scaling reduces some influence that extreme values would otherwise have but does not remove either type of concentration. Median and interquartile range scaling may also magnify tail influences if the central range of a distribution is narrow. This sensitivity is discussed further in Section 4.12 and quantified.

UMAP followed by HDBSCAN was run on twenty imputed datasets, representing ten methods and two feature representations. An embedding could represent both the geometric structure of the data and variability associated with UMAP's random initialization. To assess this variability, I ran each hyperparameter combination ten times and then selected the "best" embedding, defined as having the highest silhouette score produced during the initial application of HDBSCAN to cluster the data.

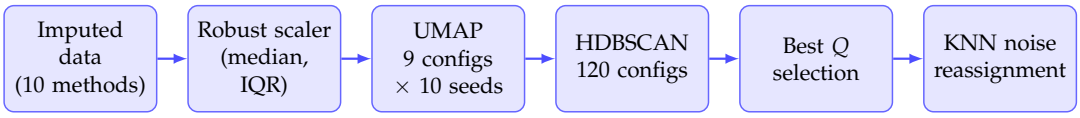


FIGURE 3.1: This is an end-to-end clustering pipeline. Once the data have been imputed, they first pass through a robust scaler. Next, after being scaled, the imputed data pass into a UMAP hyperparameter grid with 10 random seeds for each of nine configurations, and the resulting embeddings that yield the highest silhouette value are retained. HDBSCAN then does the clustering in a 120-configuration sweep, based upon the composite quality measure $Q = \text{silhouette} \times (1 - \text{noise fraction})$, where all boundary noise points are assigned back to their respective cluster(s) via k-nearest neighbour. This entire process is run over each of the 10 possible imputation outcomes.

3.5.1 UMAP Parameter Grid

I utilized Euclidean distance for the similarity measurement when comparing the points. I selected `min_dist`=0.0 to encourage separation among clusters (dense areas) in the embeddings.

TABLE 3.3: UMAP hyperparameter grid. All combinations of these values were evaluated, giving 9 configurations.

Parameter	Values
`n_neighbors`	15, 30, 50
`min_dist`	0.0
`n_components`	3, 5, 8

3.5.2 HDBSCAN Parameter Grid

Each UMAP embedding was associated with 120 different configurations for HDBSCAN. Each configuration generated a candidate cluster solution. I utilized a compound quality score to select from these candidate cluster solutions. The quality score penalized those candidate cluster solutions that were able to achieve a high silhouette value but had many observations classified as noise:

$$Q = \text{silhouette} \times (1 - \text{noise proportion}) \tag{3.4}$$

TABLE 3.4: HDBSCAN hyperparameter grid. All listed value combinations were evaluated, giving 120 configurations for each UMAP embedding.

Parameter	Values
`min_cluster_size`	500, 1000, 1500, 2000, 3000
`min_samples`	5, 10, 25, 50
`cluster_selection_method`	eom, leaf
`cluster_selection_epsilon`	0.0, 0.1, 0.5

3.5.3 Sweep Procedure

All of the imputed datasets underwent the same process flow.

Firstly, data were scaled via a robust scaler.

Secondly, for each combination of hyperparameters used for UMAP, ten random seeds produced ten embeddings.

I chose the embedding from each combination which had the highest silhouette coefficient. These runs tested sensitivity to random seed initialization and were not cross-validation splits.

Thirdly, I ran 120 different HDBSCAN parameter combinations and selected the one with the best quality score Q. That was a search over hyperparameter space rather than a 120-fold cross validation.

Fourthly, observations originally assigned to noise class were moved based on k-nearest neighbour similarity in the UMAP space ($k = 10$). The exported assignments had 100% coverage. The reference fitting process allowed no greater than a 10% reduction in silhouette when assigning labels; however, this provision was not applied during the subsequent method by method export of noise assignments.

3.5.4 Worked Example

To illustrate, let us consider an assessment with twelve observed retained test results and missing scores on the timed attention task and delayed verbal recall. Using

MICE, in the historical analysis, these two missing values are filled-in. Six composite scores are generated through direction alignment, standardization and domain weighting from these scores. Afterward the scores undergo robust scaling and are transformed using UMAP. Density-based grouping occurs next via HDBSCAN. Noise points that are eligible for reassignment receive a separate KNN reassignment as described earlier. Labels describing severity based on this cohort of patients include Above-Average, Near-Normal and Global Impairment. H2 represents a separate fixed-reference diagnostic; H2 does not assign a clinical confidence rating to this example. CogDash's New Profile page applies the saved patient-level preprocessing and either the two- or three-group k-means model in Section 3.8, reporting distances to the fitted centers.

3.6 Hypothesis Definitions

H1 (Stability and Discreteness of Cognitive Clusters). Most likely, for many imputation methods, the UMAP + HDBSCAN pipeline will define at least two stable cognitive clusters. Criteria for evaluating the stability of each cluster include: First, that the top solution produced by at least eight of the ten imputation methods has an average silhouette value of at least 0.40; Second, that no more than 30% of assessments were assigned to "noise" prior to their assignment to one of the defined clusters. These criteria alone do not establish that the partitions generated in the selected feature space correspond to distinct cognitive phenotypes. To evaluate discreteness beyond profile shape and to assess sensitivity to the number of clusters found, PCA and separate assessment in the original six domain space will be performed. Cluster-count sensitivity will also be examined.

H2 (Robustness Across Imputation Techniques). Moderate to high agreement between imputation methods is expected with respect to grouping when the geometric representations and clustering models are held constant. The principal means of discovery will continue to be UMAP + HDBSCAN. For H2, the secondary means of measuring agreement will involve applying a k-means labeling rule that was developed using data from the MICE embedding without modification to the other imputed datasets embedded into the same geometry. Agreement will be indicated where the median ARI is greater than 0.50 across all 45 pairs of imputation methods. NMI will provide supplementary evidence for agreement. Some latitude exists in choosing a threshold. The chosen threshold represents substantial overlap relative to chance and therefore indicates that cluster assignments are robust within a given reference framework. It does not require or claim that independently refitted UMAP + HDBSCAN pipelines recover the same number of tiers.

H3 (Relationship Between Clinical Unit and Cluster Membership). I anticipate that there is a connection between the clinical units and cluster membership.

To assess whether my anticipation will be confirmed, I plan on developing contingency tables, and then evaluating chi-squared tests of independence for each imputed dataset. If the results confirm a connection between clinical units and cluster membership, it will be indicated as being a statistically significant relationship at $\alpha = 0.05$ post-Bonferroni adjustments, along with Cramér's V greater than 0.10. As an additional sensitivity evaluation to account for possible confounding variables, I also plan to conduct a Cochran-Mantel-Haenszel test that is conditional upon an appropriate covariate selection.

H4 (Clustering at Domain Level versus Variable Level). I feel that grouping the cognitive attributes into their respective domains will assist in creating clusters that are easier to understand than when using the single attribute levels. Furthermore, I believe that the domain level clusters will demonstrate either comparable or improved separations compared to those found through the variable level clusters. H4 will be supported if at least two of the three metrics demonstrate that the domain level clusters were equivalent to or better than the variable level clusters. Those metrics that I plan on utilizing include silhouette score, mean VIF and condition number. Silhouette score indicates larger values represent better cluster separation. Lower values of VIF and condition numbers indicate lower degrees of multicollinearity issues and better numerical conditioning. Agreement across imputations was already assessed in H2; comparison of feature representations belongs to H4.

3.7 Statistical Tests and Validation

3.7.1 Statistical Conventions

The basic method for determining how well the research results support the statistical hypotheses is through traditional statistical procedures. Each assessment compares the data with a null hypothesis H_0, representing a no-effect baseline. The p-value is the probability of a test statistic at least as extreme as the one observed, assuming H_0 is true. Smaller p-values represent greater evidence of H_0 being incorrect. A predetermined level of $\alpha = 0.05$ is used as the criterion for determining when to reject H_0. An additional aspect of this analysis includes creating confidence intervals (CIs). If you were to randomly sample multiple times from the same population using the same methods, 95% of those CIs would contain the true population parameter.

Bootstrapping generates new datasets using random sampling with replacement from your original dataset. These new datasets allow you to create a simulation of what your sampling distribution of a statistic might look like. From these bootstrapped datasets, you can determine the 2.5th percentile and 97.5th percentile of your distributions to create a bootstrap confidence interval. The z-score describes

how many standard deviations away from the mean of a specific reference distribution a given score is. With the standardized scoring on the instruments used in this study, we are able to compare scores from different tests.

Cohen provided guidelines regarding the size of effects based upon the standardized difference (d): $d = 0.20$ represented a small effect, $d = 0.50$ was a medium effect, and $d = 0.80$ represented a large effect. In chi-square analyses, Cramér's V provides information about the strength of association. As such, I am utilizing descriptive standards to help interpret the magnitude of association; i.e., $V = 0.10$ represented a small effect, $V = 0.30$ represented a moderate effect, and $V = 0.50$ represented a large effect.

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \qquad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}, \tag{3.5}$$

3.7.2 Tests and Validation Procedures

Bootstrap Stability. I assess the bootstrap stability for each imputation method at the assessment level using $B = 50$ bootstrap samples. This resampling is done per row, so that all assessments from one subject are treated as if they were separate rows. On each of these bootstrap samples, I run HDBSCAN on the pre-computed UMAP embeddings. I compute the pre-KNN noise ratio and the median silhouette value (Equation 2.17) for each of the bootstrap samples. The pre-KNN noise ratio represents those cases where there was no cluster assigned at this stage:

$$\text{noise} = \frac{1}{n}\big|\{\, i : \text{label}(i) = \text{noise} \,\}\big|, \tag{3.6}$$

The pre-KNN noise ratio is used in the quality function Q (Equation 3.4). To meet H1, it requires that the median silhouette be greater than or equal to 0.40 and that the pre-KNN noise ratio be less than or equal to 0.30. In addition to meeting the two above mentioned criteria, at least eight of the ten methods assessed must meet them. I also apply Hennig's matching procedure to the pre-KNN labels to assess each cluster's stability, using a Jaccard threshold of 0.7 [Hennig, 2007].

Pairwise Agreement. My secondary sensitivity analysis utilized a common reference pipeline. The primary tiers were derived using UMAP + HDBSCAN. There were three types of models: 1) One StandardScaler, 2) one UMAP model and 3) one k-means model with $k = 3$. These models were fit using MICE-imputed domain scores. Each set of fitted objects was then applied to every other imputation. Imputed values changed; however, fitted objects remained constant. This allowed me to determine if the assessments were located within the same regions of a common embedding and partition. It does not allow comparisons of the geometric structure, density hierarchies, or number of tiers produced when independently fitting UMAP

+ HDBSCAN. Heat maps containing pairwise ARI (Equation 2.18) and NMI (Equation 2.19) across imputation methods are presented in Chapter 4. Agreement for each assessment is defined as follows:

$$\text{conf}(i) = \max_{\ell} \frac{1}{M} \sum_{m=1}^{M} \mathbf{1}[\text{label}_m(i) = \ell], \tag{3.7}$$

High-consensus assessments are those that received an assignment to the most frequently occurring cluster by at least $0.8M = 0.8 \times 10 = 8$ methods. This ratio is not a calibrated clinical probability.

Chi-Square Test, Cramér's V and CMH Test. A contingency table provides a tabular representation of the joint distribution of two categorical variables, e.g., cluster assignments and clinical units. Pearson's chi-square test compares the observed counts O_{ij} with the expected counts E_{ij} under the assumption of independence:

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \qquad E_{ij} = \frac{O_{i\cdot}\, O_{\cdot j}}{n}, \tag{3.8}$$

In this case, $O_{i\cdot}$ and $O_{\cdot j}$ are the sum of each row and each column. For the case of r rows and c columns, the test statistic will be assessed using a chi-square distribution with $(r-1)(c-1)$ degrees of freedom to get the p-value. Cramér's V scales the degree of association from zero to one, standardizing the chi-square value for sample size:

$$V = \sqrt{\frac{\chi^2}{n\,(k-1)}}, \qquad k = \min(r, c), \tag{3.9}$$

The Cochran-Mantel-Haenszel (CMH) Test extends the association analysis to multi-stratum data, such as sub-groups based upon diagnoses. The CMH test determines if the statistical significance of the association between clinical unit and cluster assignment is maintained after stratification by a relevant covariate:

$$\chi^2_{\text{CMH}} = \frac{\left(\sum_h (a_h - \mathbb{E}[a_h])\right)^2}{\sum_h \text{Var}(a_h)}. \tag{3.10}$$

Patient-Grouped Multinomial Sensitivity Analysis. Since associations do not establish predictive performance, I compared two regularized multinomial logistic models for detailed clinical-unit assignment. The first model, which is the base model, uses the broad diagnostic categories provided in the clinical unit look-up file. The second model, or augmented model, includes cognitive-severity tier. Clinical units that represented less than 50 unique patients were combined to create a new "other units" category, creating 17 outcome classes. I used five-fold stratified group cross-validation so that all assessments from the same patient would always fall into a single fold. My held-out multinomial log-loss was my primary performance metric. Accuracy and macro F1 metrics were evaluated descriptively. I bootstrapped

the difference in out-of-fold log loss 1,000 times by patient. As both the broad diagnostic category and the detailed clinical unit represent the same source, this model represents a sensitivity analysis of additional classification performance rather than independent external validation.

Variance Inflation Factor. VIF measures the extent of multicollinearity in regression models. For a variable j, it is calculated as follows:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}, \tag{3.11}$$

Here, R_j^2 is the coefficient of determination from regressing j on all the other predictors. A VIF greater than 10 warrants concern about multicollinearity, with implications for interpreting individual variables and the subsequent domain-level analyses.

3.7.3 Implementation

Historical analyses were run using Python 3.13. An additional pinned Python 3.12 environment was created to allow for a patient-level extension. The methods of imputation come from a variety of packages including `scikit-learn` for KNN and NMF, `fancyimpute` for SoftImpute and expectation maximization (EM), `miceforest` for MICE and predictive mean matching (PMM), and `missforest` for MissForest. Denoising Autoencoder (DAE) and Variational Autoencoder (VAE) were implemented in `PyTorch`. UMAP and HDBSCAN were utilized through the use of the `umap-learn` library and the `hdbscan` library. All statistical testing was done using `scipy.stats` and `statsmodels`. Validation metrics were calculated via `scikit-learn`. The analysis code, together with the dashboard can be found at `https://github.com/zkunos/thesis_project`.

3.8 Patient-Level Clustering Extension

An exploratory follow-up to this study was conducted on October 15, 2026. The analysis explored whether the existing historical severity categories represent stable subcategories with regard to different ways the data can be represented, with regard to how the data may have been missed, and when using different sampling units. The findings from this study are supplemental to those found in H1-H4, but do not redefine those hypotheses. The supporting literature was limited to journal articles published between October 15, 2025 and October 15, 2026. Articles used in comparisons of mixture modeling of missing data, multi imputation ensemble approaches and techniques for diagnosing clusterability (in addition to comparative longitudinal analysis) included [Tong and Tortora, 2025; Tomo et al., 2026; Neville et al., 2026; Julià et al., 2026] and provided further rationale for determining if there existed structure in the measure feature space based upon distortions present in UMAP embeddings [Chen et al., 2026]. None of these articles validated a particular type of

ABI or established which methodology would produce better results than the historical methodology.

3.8.1 Cohort and Discovery–Validation Split

We identified an eligible population of 17,406 evaluations performed on 7,285 individuals, across the 14 retained cognitive tests. Identifiers for patients link repeated evaluations; there are no duplicated patient-evaluation-time combinations. No validated identifiers exist for defining episodes of injury. One patient reported differing ages at injury across assessments and was removed from trajectory modeling based on their evaluation history.

Patient assessments were ordered by time post-injury, and each patient's first eligible evaluation was used in the cross-sectional aspect of this study. When determining the order of assessments for multiple evaluations within a single patient, source row ordering is used to break ties deterministically. In a fixed random split using seed #20261015, 5,828 patients were randomly assigned to the training set (discovery) and 1,457 patients were randomly assigned to the testing set (validation). Since each patient provides only one index profile, no patient appears in both groups. This allows us to estimate structure in patient index profiles while we describe assessment state across follow-up in the historical analysis. It has also already been noted that the 14 feature variables chosen for use in this research were those same as the feature variables used in our previous historical analyses for the entire cohort of patients. Thus, the division as described does not represent an independent validation of how we chose these feature variables.

Eligibility required at least one observed retained test. A separate sensitivity analysis evaluated the impact of sparse measurements of patient assessment profiles. Specifically, it used index evaluations where at least seven of the 14 retained tests had been measured.

3.8.2 Shared Preprocessing and Feature Representations

Data in discovery datasets establish the 1st and 99th percentiles, means and standard deviations for each test. Values are clipped at those percentile bounds, and timed `ATMTA` scores are sign-inverted. Standardization is employed to create larger values associated with higher achievement. This transformation leaves missing entries unchanged. For completed-data branches, a bounded Bayesian chained-equations imputer was fitted on discovery patients and used to impute missing values for validation patients. An iterative, deterministic reference model was created using 15 iterations. An additional sensitivity study utilizing 50 iterations was conducted to assess convergence of the iterative algorithm. Note that this imputation method differs from the historical `miceforest` implementation and has its own modeling assumptions.

The completed standardized tests were grouped by six domains based on the frequency of each test's administration. Each domain's weight is determined by the frequency of each test being administered to the discovery dataset; however, the weights have been normalized within each domain. Domain median and interquartile range (robust scaling) derived from the discovery dataset were utilized to calculate the reference distance representation. The comparator options include either the 14 completed tests or the 5 non-orientation domains. Following removal of the first principal component derived from the discovery patient data, a five-dimensional PCA basis was developed. Residual representation represents a diagnostic tool for identifying potential shape differences amongst profile groups. Therefore, residual representation may remove valid signals as well as common variability.

After examining discovery-set variances, an additional scaling sensitivity was introduced in which each domain has a variance of one. This sensitivity is designed to evaluate the influence of domains having "ceiling" issues which tend to have very narrow ranges around their midpoints. By evaluating identical label sets in both geometric models, we can separate differences in how the data are divided from differences in how silhouette is calculated. The transformations applied to the validation patients remain unchanged from those originally estimated in the discovery data.

In terms of analyzing each patient individually, the patient level analysis utilizes clipping, direction alignment and test standardization before the values are then combined. Previous supplementary analyses utilized multiple imputation and complete case comparison methodologies to aggregate raw tests. Consequently, the prior results that were provided as part of this study used a different preprocessing methodology than the patient-level extension. Controlled comparisons using identical completed test values assess the impact of aggregation without recalculating all the previous H1-H4 statistics.

3.8.3 Original-Space and Incomplete-Data Models

The clustering techniques k-means and PAM (sampled partitioning around medoids) were evaluated for values of $k = 2, \ldots, 8$. PAM utilizes CLARA-style sampling on three sets of 400 discovery patient data, followed by a BUILD/SWAP optimization within each sample. Additionally, the Gaussian Mixture Model was evaluated for $k = 1, \ldots, 8$. Both full and tied covariance matrices were used. HDBSCAN was run against the data for two sets of parameters: (a) minimum cluster size of either 100 or 300; and (b) `min_samples`=20. For the comparison utilizing UMAP, three-dimensional representations were employed, along with 15 or 50 nearest neighbor values with a `min_dist`=0 before applying the same HDBSCAN parameters as above. In terms of assigning validation patients to clusters, this was performed using the fitted centers, medoids, mixture probabilities, or UMAP transformation and HDBSCAN approximate prediction.

Models to fit each representation/method were selected based upon the highest discovery silhouette for partition/density methods and lowest discovery BIC for Gaussian mixtures. When applicable, fits had to converge and every group had to contain at least two percent of the discovery patients. Silhouettes were computed in the respective original representation (including UMAP-derived labels). Up to 2,000 randomly sampled observation discovery scores were used when computing discovery scores. All available assignments were used when calculating validation scores. Noise was excluded from silhouette, and coverage was reported separately. Scores from different feature geometries are not a common performance scale. A silhouette search for $k \geq 2$ does not determine whether there exists a group relative to an existing one population model.

The incomplete-data mixture models were fitted to 14 transformed test results using expectation maximization (EM). Missing coordinates were integrated out in the observed-data likelihood. There were three restricted families used: (a) Gaussian; (b) t-distributions with df = 4 & 10; (c) contaminated normal with a 90% normal core and tenfold covariance inflation for the remaining 10%. Each of the above mentioned families were estimated with $k = 1, \ldots, 8$, using two different starting conditions, with a 120-iteration limit or convergence when the change in log likelihood per observation was below 10^{-5}, and an eigenvalue floor of 0.02 to the dispersion matrix. Due to scale matrices being used within the t family estimation process, the eigenvalue floor also served as a scale-matrix floor. Mean Imputation was the only method used for developing starting values. Verification of these implementations has been done through computational verification. These are simplified versions of the full *MixtureMissing* package [Tong and Tortora, 2025] and lack some functionality.

BIC, entropy-penalized ICL, convergence, validation observed-data log likelihood and posterior membership probabilities were recorded for each fit. Selected fits had to converge and assign at least 2% discovery membership to each component. Likelihood measures are compared only among fits sharing the same transformation into the 14-dimensional space. Because continuous distributions are being used to approximate discrete/bounded scores, any additional groups could represent ceiling effects, skewness, or tailing rather than separate subpopulations of patients.

3.8.4 Independent Refitting, Consensus and Cluster Tendency

Twenty stochastic imputed datasets were generated from bounded Bayesian posterior draws, using ten chained-equations iterations per dataset. K-means and sampled PAM were fitted separately with fixed $k = 2, 3, 4$, refitting domain scaling for each draw. Ensemble PAM used two samples of 300 discovery patients. Each assignment vector contained the same 1,457 validation patients. Agreement across draws was measured by mean pairwise ARI and empirical 5th-95th percentiles. These are not confidence intervals because partition pairs share fits. A separate discovery-silhouette k-means screen assessed $k = 2, \ldots, 8$ in each run.

Fifty runs used 80% subsamples of discovery patients to fit preprocessing, imputation, scaling, and the fixed k=3 group k-means. Fifty additional runs were made where the clustering seed varied while the input vector was kept constant. The patient sampling variability was assessed by computing the held-out ARI and matched per-cluster Jaccard indices to differentiate it from initial condition variability. Additional sensitivity checks assessed the effects of mean versus KNN imputation, no clipping, stricter availability requirements for tests, and independent fitting of the validation partitions.

Finally, I compared average-linkage co-assignment, greedy mean-NMI, and generic NMF as three approaches to summarize the descriptive properties of held-out assignment vectors using methods similar to [Tomo et al., 2026]. Although these are the native implementations of those methods, they are not an exact replication of that paper. Additionally, consensus regarding the validation assignments does not constitute a predictive model which has been separately validated. Collapsed groups and nonconvergence are reported.

The results from the PCA/DIP diagnostic analysis were then compared to a matched-size screening set of 1,000 discovery patients and 100 separate reference samples that were each constructed by sampling from a single Gaussian copula. The empirical discrete marginal distributions as well as the missingness joint mask created through the application of resampling techniques were preserved for each. Additionally, each of these 100 reference datasets were subjected to the exact same processing procedures used on the initial dataset. Specifically, this included: preprocessing; estimation of missing data; and K-Means Clustering Analysis with varying cluster counts ($k = 2, \ldots, 8$). The tail fraction comparisons include both the observed Max Silhouette and PC1 Dip Statistics when compared to those present within the reference distribution(s). The method of comparison described above is known as "descriptive fitted-reference calibration". The simulation missingness masks are generated independently of the simulated data and selection across all model families is not repeated. In addition, there is a two group positive control check to check basic recovery. However, neither of the screens or the projection based modality test identify clinical subtypes [Neville et al., 2026].

3.8.5 Patient Trajectories and Reproducibility

To qualify for the longitudinal sample, a patient had to meet all of the following requirements. There had to be at least three visits with seven or more observed tests and at least three distinct time points. The interval between the first and last eligible observation could be no shorter than 0.25 years and no longer than five years. Recorded age at injury also had to be consistent across assessments.

I developed a longitudinal cognitive score using the frozen index-discovery preprocessing, imputer and robust-space PC1. The score was oriented towards better

performance and divided by its index-discovery standard deviation. Time was measured in years since the first eligible time point. Each patient kept their original discovery or validation status throughout the longitudinal analysis.

I modelled the data using Gaussian linear trajectory mixture models. The patients' random intercepts and slopes were integrated into the models. Models varying in number of classes (from 1 to 4) were fit. Random-effects covariance and residual variance were shared across all of the classes. For each model, three different sets of initial values were tested and up to 300 EM iterations were allowed. Only models that converged as well as had an average posterior probability of at least 0.05 in each of their respective classes could be evaluated with respect to discovery BIC. Therefore, any model that failed to converge during its estimation procedure or had an average posterior probability < 0.05 in any particular class would be eliminated from consideration. To provide a point of comparison, we also ran a one-class model in which there was a continuous random effect. Additionally, patient subsampling was conditional on selecting more than one class. Rather than determining how much of the identified pattern(s) could be due to underlying causal mechanism(s), we wanted to determine what other observed patterns of interest existed. We did not run analyses of nonlinear trajectories. Also, we did not utilize the recently developed multivariate contaminated longitudinal model. Recent studies of longitudinal data motivate separating initial value from rate of change [Julià et al., 2026].

All analyses were performed using Python 3.12.14 with fixed version numbers for all dependencies, reproducible random seed states and as previously stated, model grids. Eight numerical tests evaluated the reference likelihoods, recovery of missing data, monotonicity of EM log-likelihood, preprocessing directionality, selection of the medoids, integration over trajectory likelihoods and evaluation of the overall ensemble objective function. No independent rehabilitation outcome was available, so predictive clinical utility remains untested.

Chapter 4

Results

The initial portion of Chapter 4 reports the assessment-level analysis, while Section 4.12 includes a patient-level extension. Each additional comparison for supplemental analysis using both complete cases and multiple imputation was made using raw test scores. Domain scores were created in the extension by first aligning test direction and then standardizing the tests. The preprocessing specifications differ, so these results are not directly comparable.

4.1 Exploratory Data Analysis

In addition to what has been stated before in Chapter 3, when using tier-1 filtering and removing all assessments that do not include some type of cognitive measure, there were 17,406 assessment records for 7,285 patients. The resulting data matrix contains 14 cognitive variables that fall into 6 different domains. This is listed out in a table format in Table 4.1 showing the percent missing for each variable.

There is considerable variation in the amount of missing information. The percentages of missingness vary from 0.7% on the Orientation Items to 27.8% on the ATMTA. Were only complete cases to be used, 46.9% of the assessments would be excluded from analysis with a resultant potential for biased selection [Rubin, 1976]. To prevent loss of such a significant number of assessment samples in the eligible population, I decided to perform an imputation.

Although a co-occurring heatmap (not included) did show that there are times when multiple variables are missing at the same time, it does not provide evidence of missing-at-random mechanisms [Rubin, 1976; Little and Rubin, 2019]. These dependencies suggest that conditional imputation may be superior to naive substitution. Generally speaking, correlation coefficients among the various variables in a given domain are relatively high when compared to correlation coefficients among variables in other domains. High shared variances among the various cognitive measures are compatible with a general cognitive-ability component [Lezak et al., 2012] which the subsequent clustering analyses examine further.

The graphical representation of the percentage of missing values by cognitive measure is shown in decreasing order in Figure 4.1. Routine screening measures

TABLE 4.1: Missingness summary for the 14 eligible neuropsychological variables after Tier 1 filtering. The final row reports the number of complete cases (rows with no missing values across all 14 variables).

Variable	Domain	*N* Observed	*N* Missing	Missing %
OTEMPS	Orientation	17,286	120	0.7
OPERSONA	Orientation	17,284	122	0.7
OESPAI	Orientation	17,282	124	0.7
ASPAN	Attention	15,611	1,795	10.3
MRAVLT075	Memory	15,523	1,883	10.8
MDIGITS	Memory	15,373	2,033	11.7
FEPMR	Executive Function	14,682	2,724	15.6
LCOMPRENSIOTB	Language	14,686	2,720	15.6
LDENOMINACIOTB	Language	14,677	2,729	15.7
LREPETICIOTB	Language	14,634	2,772	15.9
MRAVLT015	Memory	14,281	3,125	18.0
MRAVLT015R	Memory	14,199	3,207	18.4
VPIMATGES	Visuoperception	12,992	4,414	25.4
ATMTA	Attention	12,575	4,831	27.8
Complete cases (all 14 variables observed)				**9,245 (53.1%)**

generally have less missing data than special purpose assessments. This is consistent with their common utilization in clinical practice [Lezak et al., 2012].

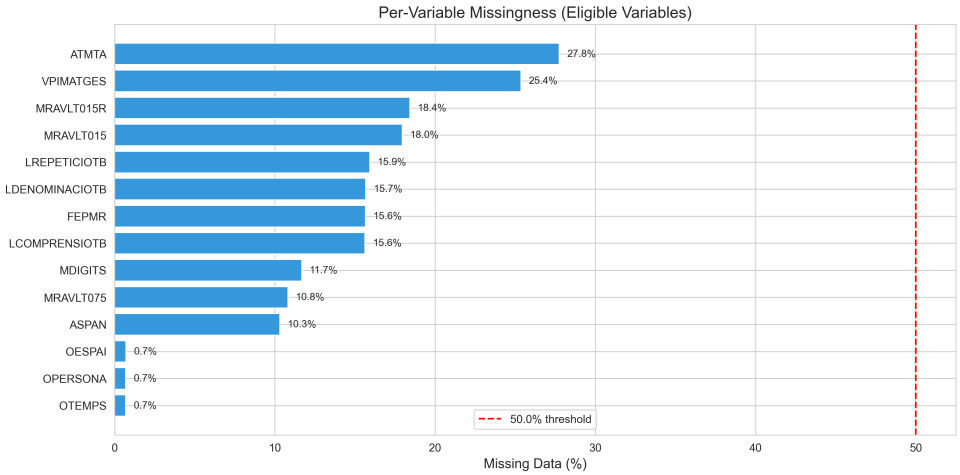


FIGURE 4.1: Missingness bar plot for all cognitive variables, ordered by missingness rates. Missingness rates were higher among the specialised assessments compared to the routine screening assessments.

4.2 Imputation Results

Prior to determining how well each cluster model performs on each data set, I first compared each of the ten different imputation models (as presented in Figures 4.2 and 4.3). The reason that mean imputation differs from the remaining 9 methods is that mean imputation loses variance when it replaces missing values with the mean [Rubin, 1987], while the remaining 9 methods provide closer approximations of the original marginal distributions. In this comparison, KNN and PMM appear

to maintain more of the visually discernible discrete structure than do methods like MICE, MissForest and EM, which are producing smooth continuous reconstructions.

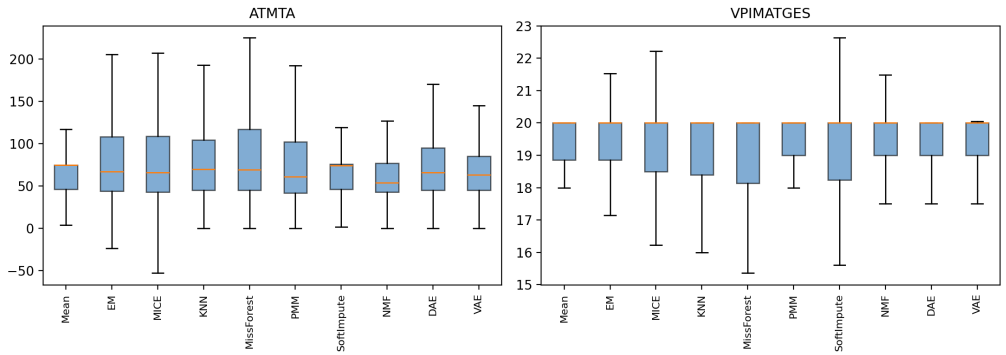


FIGURE 4.2: Comparison of distributions for each of the 10 different imputation methods for ATMTA and VPIMATGES. Mean imputation reduces variance whereas model-based methods maintain the distributional characteristics of the original data.

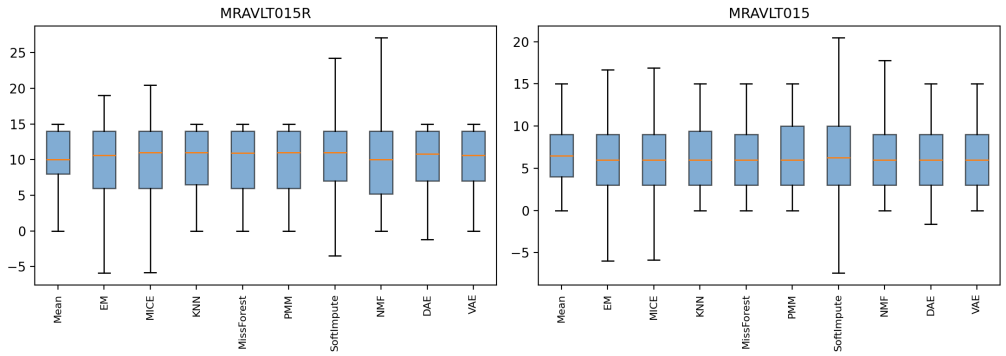


FIGURE 4.3: Comparison of distributions for MRAVLT015R and MRAVLT015. Deep learning models (DAE, VAE) closely approximate model-based methods.

4.3 Baseline and Primary Clustering

First, I ran K-means and Gaussian mixture models as baseline approaches to identify the number of groups (k) from $k = 2$ to $k = 6$. Neither method revealed a clearly defined optimal number of groups. There was no distinct "elbow" identified by the K-means approach. Additionally, the maximum silhouette score for all values of k was found at $k = 2$, with scores decreasing as the number of groups increased. These results motivated an investigation of density; they were not evidence that natural cognitive groups existed.

I chose HDBSCAN next due to its ability to locate densely packed areas which could take on irregular shapes, classify sparse points as noise, and assess cluster stability across a hierarchical tree, providing a measure of cluster consistency without needing a pre-specified number of clusters (k). This seemed suited to routine clinical data, where boundary cases may not fit cleanly into a group. As part of my analysis,

I ran UMAP using 10 random seed values for each set of parameters, and evaluated HDBSCAN using the full 120 combinations of parameters as previously described in Chapter 3. Figure 4.4 shows the selected assignments under t-SNE and UMAP projections. However, it does not serve as an independent means to determine if clusters exist and/or if those clusters are distinct.

The final UMAP + HDBSCAN parameters included `n_components`=8 and `n_neighbors`=15 for UMAP, along with `min_dist`=0.0 to allow for tight clustering within the embedding space [McInnes et al., 2018]. When utilizing HDBSCAN, I employed `min_cluster_size`=1000 and `min_samples`=50. I used `cluster_selection_method`=eom and `epsilon`=0.0. I set `min_cluster_size` to a high level to ensure that only substantive groupings would remain.

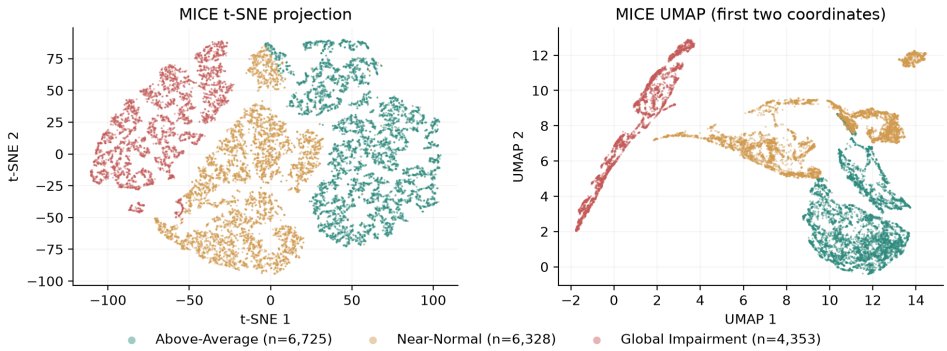


FIGURE 4.4: The same 17,406 MICE assessment assignments shown in t-SNE coordinates (left) and the first two UMAP coordinates (right). Colours identify the three final severity bands after noise reassignment; projections illustrate the fitted partition and do not establish discrete phenotypes.

4.4 Cognitive Profiles of the Clusters

I have plotted the mean cognitive-domain scores by tier as z-scores relative to the cohort mean using the radar plot. The three polygons are almost concentric, with all domains moving together; thus it appears that the primary difference among tiers lies in their global levels of functioning rather than their pattern of deficits or strengths.

Additionally, I produced a random forest-based surrogate model which made an attempt at replicating the tier classification based on the individual domain scores. I used Gini importance to measure the degree to which each domain contributed to classifying participants into one of the three groups (tier classifications). The domains are correlated, and the tier labels were derived from the same scores. Consequently, my results indicated that the lower Gini-importance value assigned to "attention" when compared to "orientation", "visuoperception", and "language" does not indicate that attention is clinically less relevant. My findings are limited to the specific tier classification modeled by this particular fit.

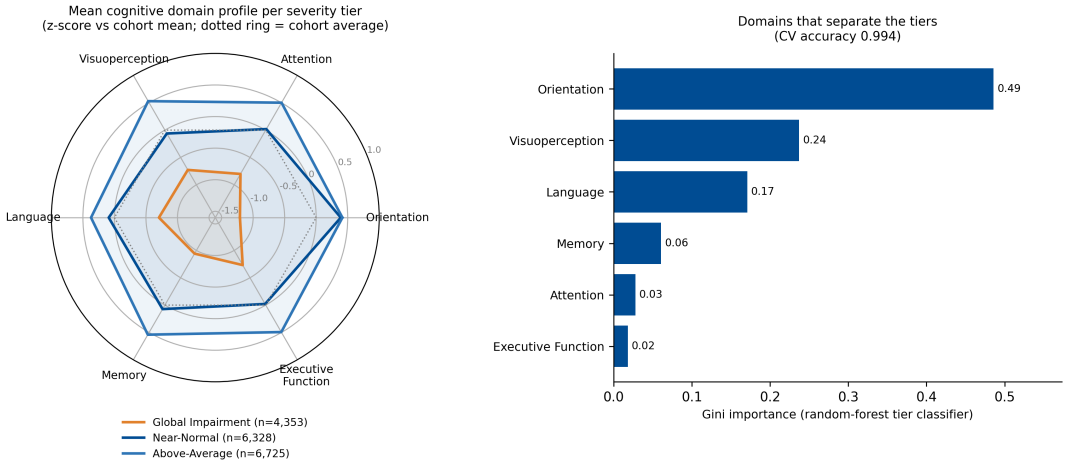


FIGURE 4.5: Left: Mean score for each cognitive domain in each tier, expressed as a z-score from the cohort mean (the dotted ring indicates the cohort average). The three polygons are approximately circular because each domain moves collectively; therefore, the tiers differ in terms of global levels rather than shapes. Right: Gini importance from a random forest surrogate that recovers tier labels from domain scores (accuracy = 0.994). Since these labels are derived from those same scores, this panel describes the decision boundaries of the partitions instead of providing independent importance measurements, and the ordering depends on inter-domain correlation (see Section 4.9). It should not be interpreted as suggesting that attention has no clinical significance.

4.5 Hypothesis 1: Stability and Discreteness

The first claim of hypothesis one is that there are replicable (reproducible) groups within the assessment data. The second claim of hypothesis one is that these subgroups are distinct (discrete) cognitive phenotypes. To assess whether or not the clusters were stable, I performed a total of 50 bootstrap samples per imputation type. For every sample I then fit HDBSCAN to the new sample of the UMAP embeddings, and calculated three metrics: silhouette, pre-KNN noise proportion, and Hennig core-Jaccard matching. I recorded the proportion of noise prior to the KNN assignment step. The metric values for each of the 50 samples created a bootstrapped distribution for the metrics for each type of imputation, as illustrated by Figure 4.6. These distributions allowed me to compare how similar/different the different types of imputations were from one another; this is summarized by the exact details in Table 4.2.

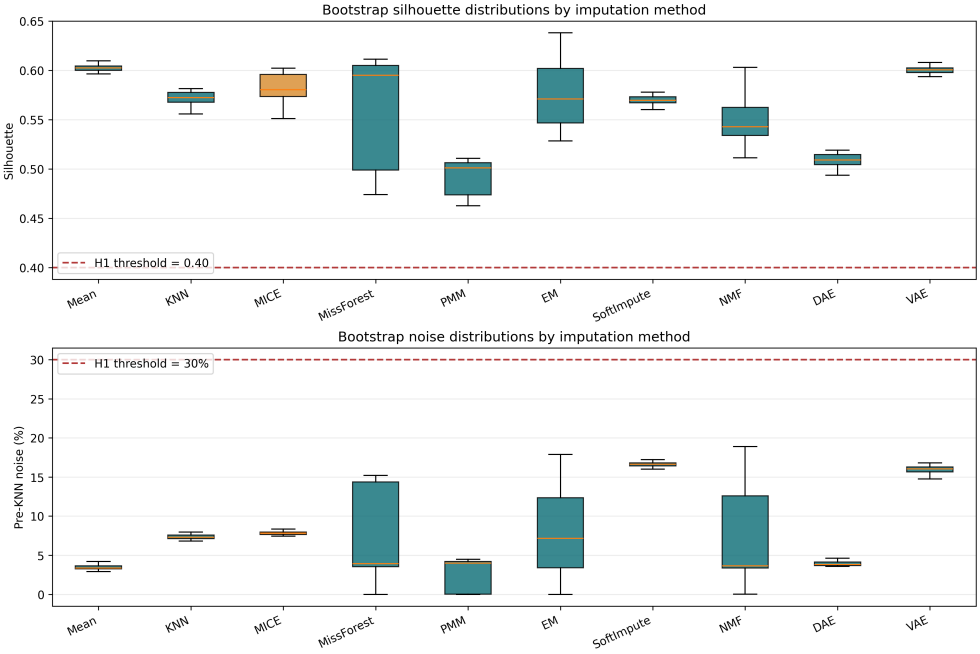


FIGURE 4.6: Bootstrap distributions of silhouette (top) and pre-KNN HDBSCAN noise proportion (bottom) across 50 assessment-level resamples for each imputation method. Dashed lines mark the prespecified H1 thresholds. All ten methods met the silhouette and noise criteria, although these values describe stability in the selected UMAP embedding and do not account for repeated assessments within patients.

TABLE 4.2: Exact per-method H1 results. Silhouette and noise are bootstrap medians before KNN reassignment. Overall Jaccard is the mean matched-cluster value; the final column gives the range across clusters within each method.

Method	k	**Silhouette**	**Noise %**	**Jaccard**	**Cluster range**
Mean	4	0.602	3.4	0.630	0.627–0.633
KNN	4	0.572	7.3	0.600	0.543–0.632
MICE	3	0.581	7.8	0.573	0.467–0.630
MissForest	3	0.595	3.9	0.570	0.536–0.631
PMM	4	0.501	4.0	0.615	0.580–0.631
EM	4	0.571	7.1	0.509	0.383–0.630
SoftImpute	4	0.570	16.6	0.610	0.586–0.631
NMF	3	0.543	3.6	0.584	0.557–0.631
DAE	5	0.509	3.8	0.566	0.490–0.630
VAE	4	0.601	16.0	0.600	0.574–0.631

All ten imputation methods met the thresholds for silhouette and pre-KNN noise proportion. However, there was relatively weak evidence supporting strong, cluster-by-cluster reproducibility. Core Jaccard similarity scores ranged from 0.509 to 0.630 overall and individual matched-cluster similarities ranged from 0.383 to 0.633. As such, none of the methods reached the 0.70 criterion for stability. Moreover, the number of identified clusters differed across many of the methods as well

(i.e., three to five). Representations also influenced separation based upon the MICE labeling method. For instance, while selecting the UMAP embedding produced a full coverage silhouette score of 0.526, using the six-domain space with robust scaling reduced it to 0.079. Cluster profiles were concentric, and the first principal component was dominant. Overall, the results appear to demonstrate a reproducible ordering by severity, but do not provide clear-cut distinctions between natural classes. Therefore, I find H1 to be partially supported. While it seems plausible to extract severity groups/strata from data, the level of distinctiveness of the strata is still unclear. Because the bootstrap resampled assessments rather than patients, repeated assessments may have increased apparent stability and reduced uncertainty around the estimates.

4.6 Hypothesis 2: Imputation Sensitivity Under a Fixed Partition

I have also tested Hypothesis 2. This hypothesis stated that when you use the same partial data set in an identically designed imputation environment, it can reasonably be expected that the results of clusterings based upon the data sets developed through an identically defined process, will show substantial similarities under a fixed embedding and partition. Using UMAP + HDBSCAN, I found all three primary tiers in my main analyses. To conduct my second level of sensitivity analysis, I performed a StandardScaler, then a UMAP, then a K-Means labeling rule with $k = 3$ on the MICE-imputed domain scores. I did this on each of the other imputed data sets. Because I had already fit my models to the MICE-imputed domain scores, I could only vary the imputed values. Therefore, I matched up all possible methods so that I had 45 pairs of methods. My average ARI was 0.710 and my median ARI was 0.739. The lowest ARI value I obtained was 0.509 while the highest ARI value I obtained was 0.880. The mean consensus fraction was 0.929. There were at least eight out of ten methods that agreed on how they assigned members to their respective clusters in 86.2% of all assessments.

Therefore, as it relates to H2 and as stated, the assignments of members to clusters are relatively stable across many methods used to replace missing values within an identified single embedding and partitioning model. However, this does not confirm whether or not multiple independently fitted pipelines constructed via UMAP + HDBSCAN will identify similar hierarchical densities. The numbers of clusters generated for each complete-pipeline solution ranged from $k = 3$ to $k = 5$. The numbers of clusters produced in the five separate MICE complete-pipeline runs ranged from $k = 3$ to $k = 7$. This type of variability is consistent with treating a continuum as if it were discrete; however, it is not sufficient to prove this characterization.

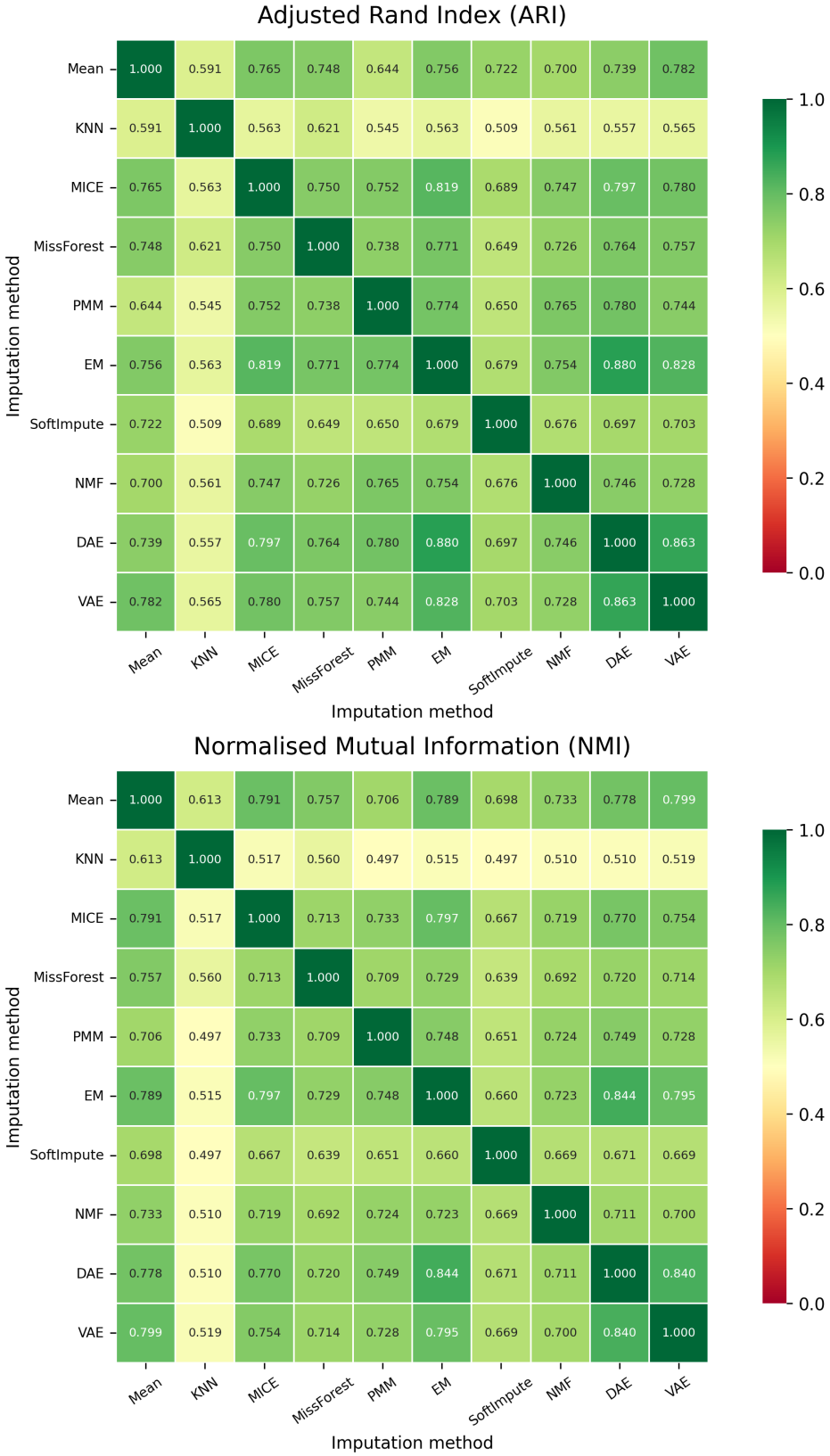


FIGURE 4.7: Adjusted Rand Index (ARI) at top and normalised mutual information (NMI) at bottom for pairwise comparisons across the 10 different imputation methods. High agreement was found throughout, although KNN showed lower agreement with the smoother model-based methods.

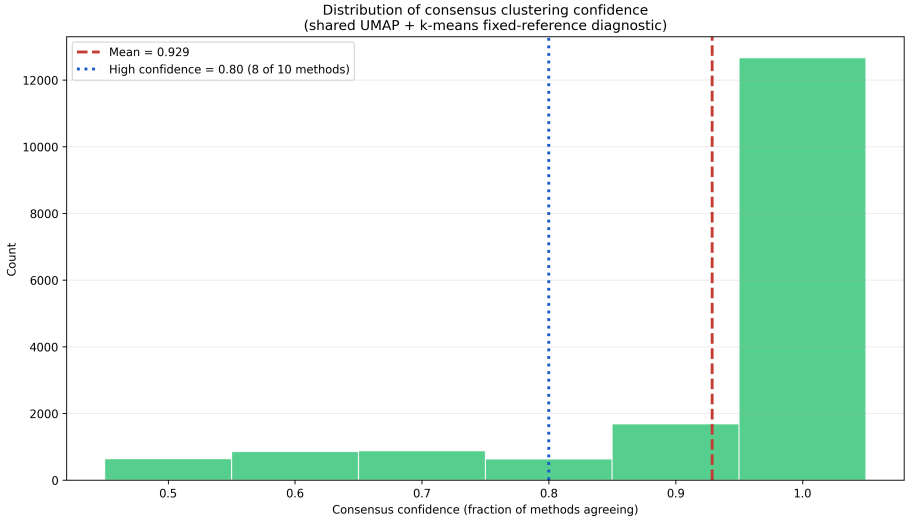


FIGURE 4.8: Histogram of consensus confidence values under the fixed-reference H2 diagnostic. The blue reference line marks the predefined high-confidence threshold of 0.80, equivalent to agreement by at least eight of the ten imputation methods. Mean confidence was 0.929, and 86.2% of assessments met this threshold.

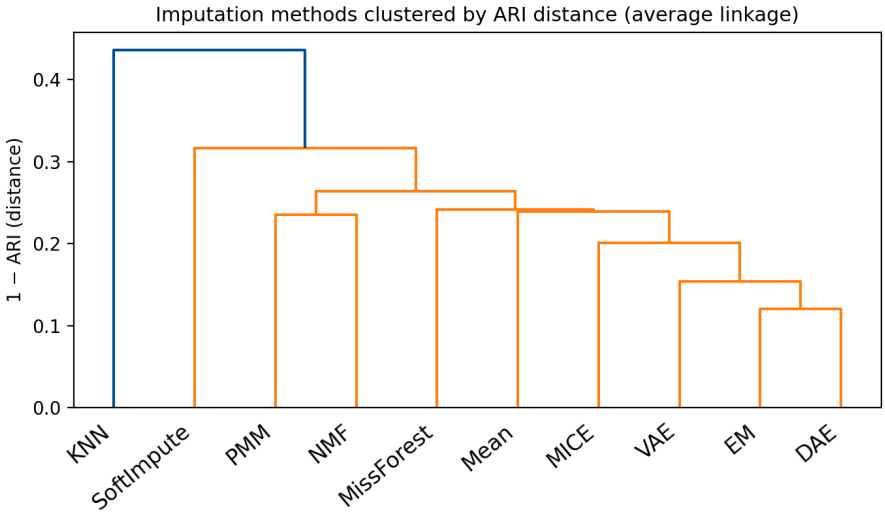


FIGURE 4.9: Dendrogram of imputation methods using average linkage clustering on ARI-based distance metrics. KNN separates itself from other branches, and tighter grouping occurs among the smoother model-based methods.

4.7 Hypothesis 3: Clinical Unit Association

Hypothesis 3 assesses if there are any associations between the cognitive-severity tier and clinical-unit assignment. A chi-square test demonstrated a statistically significant relationship between the two ($\chi^2 = 911.3$, $p \approx 4 \times 10^{-156}$), with Cramér's $V = 0.162$. This is a small effect under Cohen's descriptive guidelines, but exceeds the study's specified cutoff of 0.10. In addition, when calculating the mean Cramér's

V over five independent MICE imputations, a similar average of 0.154 was calculated. Therefore, this represents a summary of sensitivity estimates across multiple imputations rather than an inference based on Rubin's pooling rules.

A stratified sensitivity analysis also found that the association persisted even after controlling for the broad diagnostic grouping which is provided in the clinical-unit lookup. Since these diagnostic groupings were created using the same lookup data that provides the detailed unit assignments, this result should not be considered to represent any form of independent predictive validity.

Patient-grouped multinomial models gave no clear evidence that tier would improve classification of the detailed clinical units beyond the broad diagnostic grouping. Log loss for diagnosis alone was 0.6064, while log loss including tier was 0.6051, representing a decrease of 0.0013. Its 95% patient-bootstrap interval was $[-0.0006, 0.0031]$, which includes zero. The accuracy scores were 0.7772 and 0.7770, respectively, and macro-F1 scores were 0.476 and 0.480, respectively. As such, I consider H3 to be supported only insofar as it represents an association hypothesis, and does not establish added predictive value.

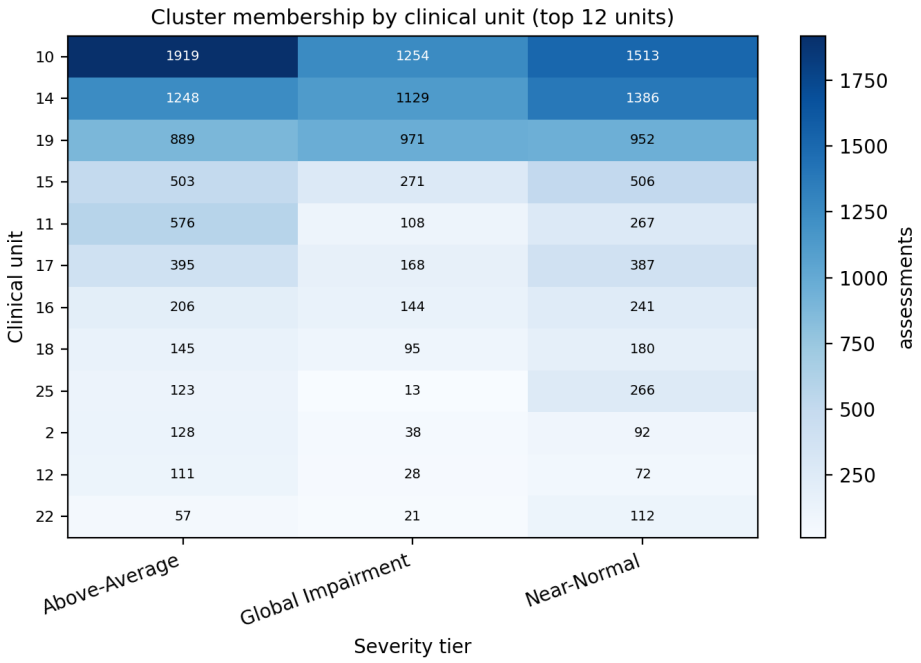


FIGURE 4.10: Heatmap of cluster assignments (columns) by clinical unit (rows). Cell colors represent frequencies. Specific tiers occur more frequently in certain units.

4.8 Hypothesis 4: Feature Engineering

The hypothesis for this comparison was that by reducing redundancy, features of an aggregate domain will be easier to interpret than the separate individual variable approach, but they should maintain separation similar to the separate variable

method. I examined cluster separation by comparing silhouette values and compared the quality of the feature spaces as represented by the condition numbers. To measure the amount of multicollinearity in each representation, I calculated the variance inflation factor (VIF).

Average silhouette improved with aggregating tests within domains; it went from 0.473 to 0.481 (see Figure 4.11). However, there was much larger improvement when examining the conditioning of the feature space. The average VIF dropped from 2.77 to 1.88, and the condition number dropped from 61.7 to 11.8. Large VIFs indicate high levels of multicollinearity among variables. As such, if there are correlated variables influencing distance calculations made by both UMAP and HDBSCAN then smaller VIFs would mean fewer redundant influences on the distances after data has been grouped into a cognitive domain. Additionally, groupings by cognitive domain allow for easier understanding and relating the results to the constructs measured by those tests [Lezak et al., 2012].

The main support for Hypothesis 4 comes from a reduction in redundancy and better numerical conditioning and interpretability. However, there is very little difference in the silhouette scores indicating no substantial improvements in clustering separation.

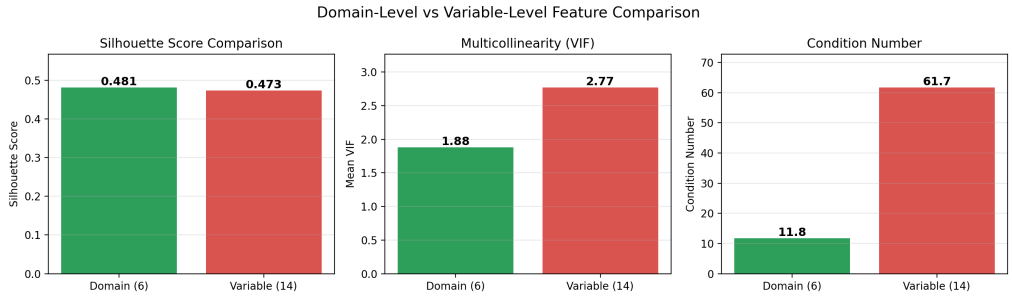


FIGURE 4.11: Comparison of domain-aggregated and variable-level features: silhouette (0.481 vs. 0.473), mean VIF (1.88 vs. 2.77), and condition number (11.8 vs. 61.7). Aggregation at the domain level produces similar separations but much less correlated, well-conditioned features.

4.9 Feature Importance Analysis

A random forest surrogate model was constructed based on the MICE-imputed domain scores using the pre-existing tier label values to define the target variable. Cross validation across five folds resulted in an overall accuracy rate of 99.4%, demonstrating that the boundary partitions derived from the same domain scores can be replicated. Although this demonstrates the reproducibility of partition boundaries, it does not independently establish natural cluster separation or predict clinical outcomes.

The Gini Importance ranking for these domains was: Orientation (0.486); Visuoperception (0.237); Language (0.171); Memory (0.061); Attention (0.028); and Executive Function (0.018). As seen in Figure 4.12, Permutation Importances provided similar ordering for the three top-ranked variables with attention ranked lowest. These results detail which variables reproduce the chosen tier labels. If multiple variables are highly correlated, there will likely be overlapping or substitutable effects among them when assessing their contribution, thus influencing both their ordered ranks [Boulesteix et al., 2012] and ultimately their relative importance to the prediction. Therefore, even if a predictor has a low surrogate importance value, it does not necessarily indicate that the predictor is clinically unimportant. Additional insights into the severity continuum are presented through the domain profile and principal component analysis in Section 4.4.

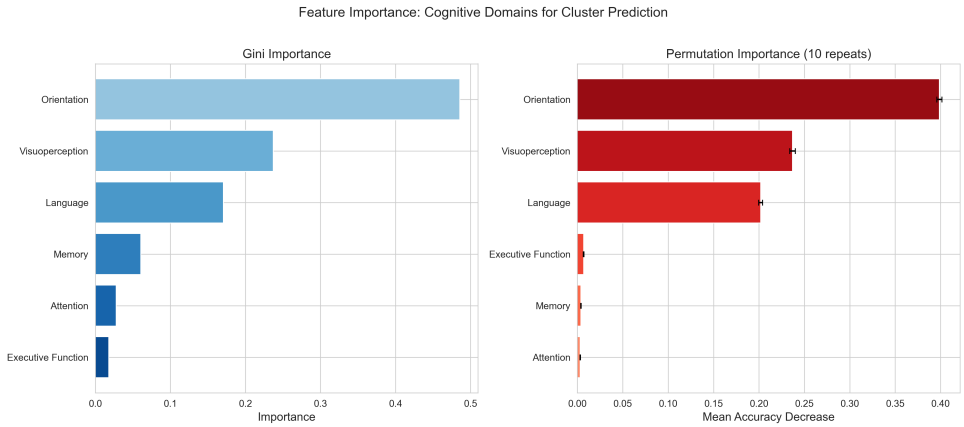


FIGURE 4.12: Random Forest surrogate trained to recover tier labels from domain scores. Left: Gini importance. Right: Permutation importance with error bars. Since labels are derived from the same scores, this panel describes the decision boundaries of the clustering partitions rather than independently evaluating their relative importance. Because the domains are strongly intercorrelated, rankings should not be taken as implying clinical insignificance of attention.

4.10 Robustness Analyses

I did additional research to see how robust the hypothesis findings were. First, I generated a baseline using a list-wise deletion approach. Next, I did an additional test called a subsample-stability test. Then, I created **Bootstrap Confidence Intervals.** Afterward, I ran independent MICE imputations and compared the results to those I had found originally. Additionally, I entered into the model each individual's progression through their tier levels (i.e., each subject's transition in tier level over time) and the amount of time (in days) since the injury. Lastly, I applied an adjustment to each of the 45 pairwise H2 statistical tests to control for testing multiple times.

Bootstrap Confidence Intervals. Using assessment-level resampling, I calculated the average ARI (Adjusted Rand Index) for H2 to be 0.710; 95% CI [0.702,

0.717] and Cramér's V for H3 to be 0.166; 95% CI [0.156, 0.175]. For H4, the silhouette gap was -0.002; 95% CI [-0.021, 0.013]. The H4 confidence interval includes zero, so it appears there is no support for increased separation of individuals based upon this additional comparison. Since these are estimates that are derived separately from the initial findings and need not reflect the same results as the headline numbers, the initial raw-test aggregation used in the supplementary feature comparisons differ from the standardized patient-level process. In addition, because this resampling treats all 17,406 assessments as if they were independent (although they actually came from 7,285 different patients), the intervals may appear tighter than warranted; they do not capture any uncertainty due to resampling patients as units.

H2 and Multiple Comparison Correction. To compare all possible combinations (forty-five total) of cluster assignments using the specified cutoff (ARI > 0.50), I used Holm-Bonferroni adjustment to control family-wise Type I error and Benjamini-Hochberg adjustment to control the false discovery rate. Using one-sided parametric bootstrap p-values for each combination, the null-hypothesis was rejected for forty-four of the forty-five combinations at both family-wise and FDR corrected significance levels. KNN–SoftImpute was the borderline case; its ARI had a mean bootstrap value of 0.509 and 95% CI of [0.497, 0.521].

Across-Imputation Sensitivity Summary. To assess variability across imputations, I created $M = 5$ independently generated MICE imputed datasets and completed the full pipeline on each dataset. Across all nonlinear statistics, I present the cross-imputation mean ($\bar{Q}$) as well as a descriptive dispersion interval for that statistic defined as $\bar{Q} \pm t_{0.975,4}\sqrt{(1 + 1/M)B}$, where B represents the sample variance of the five estimates. Since no estimation of the within-imputation variance exists, these intervals represent only the between-imputation variation and should be interpreted as such. Due to the non-linear nature of Cramér's V and silhouette, they do not represent Rubin-rule confidence intervals. Rather, they serve as sensitivity analyses providing summaries of how much the results can vary based upon different plausible sets of imputed data. The overall mean Cramér's V was 0.154 with a descriptive interval of [0.049, 0.259]. The average silhouette gap was 0.050 [0.043, 0.058]. The number of recovered clusters ranged from three to seven. Domain-level features outperformed variable-level features in each of the five runs.

Listwise Deletion Baseline. Following removal of 46.9% of the total number of assessments containing missing values, I then completed the remainder of the pipeline with the remaining 9,245 complete case assessments. This generated a two-cluster solution with a silhouette measure of 0.257. The MICE cluster solution for these same assessments demonstrated moderate correspondence with the historical solutions derived using only complete case assessments (ARI = 0.625; NMI = 0.519). However, they are not identical. As such, the use of imputation does affect how one conducts their analysis; removing 46.9% of assessments may result in differing numbers or qualities of levels/tiers as defined by this study.

Subsampling Stability. Across the twenty 80 percent random samples of assessments used in this study, on average, there was an ARI value of 0.273 among pairs of fits. The median pairwise ARI was 0.277. Fifteen fits produced a solution that consisted of two clusters, while three fits found a solution consisting of three clusters. Two fits did not produce any type of clustering solution. As such, the historical UMAP + HDBSCAN tiering assignment has considerable sensitivity based upon resampling. Results presented here relate to assessment rows and do not relate to the patient level k-means stability results discussed in Section 4.12.

Missingness Threshold Sensitivity. I repeated the full process of variable selection, domain computation, MICE imputation and UMAP + HDBSCAN clustering using five Tier-1 thresholds from 30% to 70%. As Figure 4.13 shows, at thresholds from 40% through 70%, the tier number ranged from two to three, while the silhouette scores ranged between 0.39 and 0.52. However, once a threshold of 30% was applied to the analysis, the restriction placed upon the variables resulted in a failure of the system. Therefore, the historical threshold sweep provided here will evaluate how sensitive the selected variables and resultant pipeline are and not whether the primary saved matrix was produced using a 50% row-missingness limit. The established row rule is described in Section 3.2.

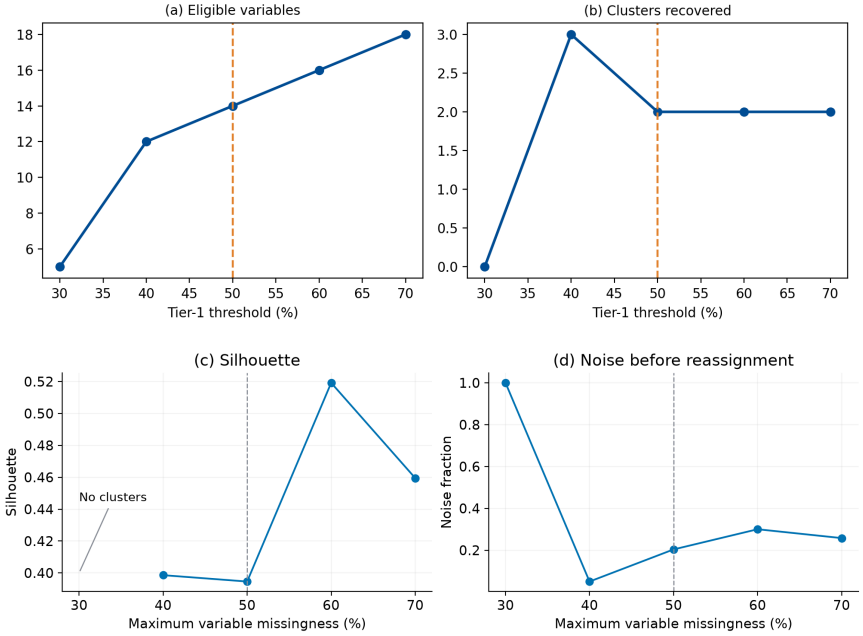


FIGURE 4.13: Effect of varying Tier 1 missingness threshold across five values (30 – 70%). Upper row: (a) Number of available variables and (b) Number of clusters identified. Lower row: (c) Silhouette score and (d) Proportion of noise. The 30% fit has no clusters, so its undefined silhouette is omitted. Vertical dashed lines indicate Tier 1 missingness threshold used in primary analyses.

Time Since Injury and Tier Transitions. The median time since injury for individuals in the Above-Average group is 214 days, 188 days for individuals in the Near-Normal group, and 172 days for those classified as Global Impairment. Due to

their nature, cross-sectional comparisons do not allow for an assessment of an individual's overall level of recovery and should never be considered as an independent measure of functional status. Out of a total of 5,206 patients who had at least one repeat assessment, there were 2,613 (or roughly 50.2%) whose tier classification did not change during any of their observed visits, while there were 2,593 (roughly 49.8%) whose tier classifications changed. Among these patients, 912 had exactly two assessments and experienced a change in tier classification: 706 of whom moved to a higher-tiered classification and 206 of whom moved to a lower-tiered classification. This represents movement within cognitive-score bands but does not establish validated functional improvement. Factors such as timing of visit; discharge; and availability of tests may have affected observed tier transitions.

Initialization and Resampling. The historical parameter search repeats UMAP initialization. The choice of a seed based upon the silhouette value of the seed as a part of model selection is not an independent validation. Above are assessments of a second (and separate) source of variation (subsampling at the assessment level). The patient-level extension reports initialization agreement and patient-subsampling agreement separately for its fitted k-means partition.

Gaussian Mixture / Latent Profile Baseline. In order to provide a statistical basis for comparison among the different classification techniques, I applied Gaussian mixture modeling (using full covariance matrices) to the imputed domain scores for $k = 2$ through 6. Specifically, this represents one way to conduct a type of latent profile analysis. However, since the analyses were based on the imputed data, they do not represent the traditional latent profile analysis using full-information maximum likelihood estimation on the raw and incomplete data. As k increases, the Bayesian information criterion (BIC) decreases from 243,249 for k=2 down to 92,606 for k=6. No Bayesian information criterion optima were identified across the various levels of k that were analyzed. While this does not establish a continuum and potentially a better fit would occur if additional components existed beyond those tested (i.e., greater than 6), the level of agreement between the tiered densities based on the methods described above and the results obtained by the mixture-modeling techniques were generally moderate (ARI ranged from 0.350 to 0.400); however, the two-component solution produced the highest value (0.45) for silhouette. Consequently, I consider the mixture-modeling procedures to produce ambiguous answers relative to the number of categories. The continuum representation has many of its characteristics dependent upon the concentric nature of the domain profiles, the dominance of the first principal component, weak original-space silhouettes and the dependency of the number of tiers on the technique used. The observed-data mixture extension in Section 4.12 fits the incomplete test matrix directly, using a wider range of restricted distributional families.

Missing-Not-at-Random Sensitivity. While there was agreement among the different imputation techniques used, it is still dependent on what each technique assumed regarding the missing data. No studies I conducted comparing the sensitivity

of all the methods revealed that the results would remain unaffected by potentially informative missingness with respect to test scores. Thus, as part of my future research, I will conduct a formally defined delta-shift sensitivity analysis that respects test direction and uses clinically justified shifts.

Assessment Eligibility. A total of 17,406 of 22,075 records contained at least one observed score among the 14 retained tests; these 17,406 records were thus analyzable. Records without measurable responses from the 14 tests in this study (i.e., n = 4,669) were excluded from analysis. The absence of retained test values in those encounters prevents direct assessment of cognitive severity. However, that does not indicate that excluding them is harmless from a clinical standpoint.

4.11 Cluster Clinical Naming

I applied domain-level z-score profiles to establish a rule-based system that would assign clinically relevant labels to density-based clusters. There are three possible labels assigned to each cluster profile.

Profiles with domain z-scores greater than +0.5 were classified as "Above Average."

Profiles with domain z-scores less than -0.5 were labeled "Global Impairment."

All remaining cluster profiles were classified as "Near Normal" or the middle category.

There were 6,725 "Above Average," 6,328 "Near Normal," and 4,353 "Global Impairment" assessments in the total assessments.

These bands have different average values but similar shape. As such, severity labels are better than phenotype names.

They provide a shorthand description of the historical assessment-level partition. Clinical utility for triage decisions has not been established for this label assignment, nor do they transfer automatically to the new patient-level groups.

In addition to providing assessment counts and domain profiles for each band, CogDash also displays the separate patient-level findings. Group numbers are specific to each fitted solution.

4.12 Patient-Level Clustering Extension: Results

The new analyses identified repeatable groupings in cognitive test scores, but their interpretation as distinct clinical phenotypes remains unresolved. These results reflect the earliest eligible assessment for each of the 7,285 patients, which included 5,828 patients who were part of the discovery cohort and 1,457 patients from the validation cohort. These assessments cannot be used as a direct comparison in terms of performance when compared to the original assessment-level UMAP analysis since these two cohorts have differences in terms of composition, representation and validation strategy.

4.12.1 Original-Space Comparisons and Cluster Count

Representative solutions generated via discovery data analysis are summarized in Table 4.3. It appears that for the reference six-domain representation, two-group k-means and PAM (Partitioning Around Medoids) algorithms were best suited. When applied directly to both of the tested minimum cluster sizes, however, HDBSCAN provided only "noise" classifications; UMAP + HDBSCAN did produce group assignments. In fact, increasing the size of the neighborhood in UMAP (i.e., changing the number from 15 to 50) resulted in changes in the number of groups identified as well as lessened separation among the groups in terms of the measured domains. As such, these results suggest that examining whether or not the measured feature representations provide adequate support for the partitions identified through the embedded representations may be useful. Nonetheless, it seems reasonable to assume additional research will need to be conducted into identifying different methods for defining the parameters of the density models.

TABLE 4.3: Selected patient-level models. Silhouettes are calculated in each row's stated representation; different geometries are not directly comparable. All listed solutions have 100% validation coverage.

Model	Representation	k	Discovery	Validation
K-means	Robust domains	2	0.666	0.649
Sampled PAM	Robust domains	2	0.553	0.540
Full Gaussian mixture	Robust domains	8	0.117	0.118
Tied Gaussian mixture	Robust domains	5	0.548	0.535
K-means	PC1 residuals	3	0.547	0.532
K-means	14 tests	2	0.406	0.399
UMAP+HDBSCAN	Robust domains	2	0.389	0.400

The k-means algorithm chose a consistent number of groups (two) in each of the twenty stochastic imputations made during the discovery-silhouette search. This was a consistent choice among $k = 2, \ldots, 8$, not a test against one population. The eight-component full Gaussian domain model also reached its maximum number of components. Neither of the other methods examined (minimum BIC at the boundary, silhouette-based selection) appeared to identify a single "natural" number of clusters.

4.12.2 Independent Imputation and Patient Stability

The two-, and three-cluster K-Means analysis had very good consistency under the constrained Bayesian imputation model, and also when using the "robust domain" representations as shown in Table 4.4. The average ARI among the pairs of clusters generated from the fixed 3-cluster patient sub-samples was 0.968. When the clustering seed was changed while holding the input constant the average ARI was 1.000. The average Jaccard match value over the three cluster assignments was 0.994, 0.959 and 0.947 respectively. These numbers demonstrate repeatability of the grouping

across this preprocessing pipeline and do not indicate that there is a clinically meaningful distinction between these groupings.

TABLE 4.4: Pairwise agreement on the same validation patients after independent fits across 20 stochastic imputations. Percentiles describe dependent pairwise comparisons and are not confidence intervals.

Method	k	Mean ARI	5th–95th percentiles
K-means	2	0.959	0.944–0.974
K-means	3	0.935	0.917–0.951
K-means	4	0.738	0.627–0.823
Sampled PAM	2	0.909	0.851–0.959
Sampled PAM	3	0.504	0.281–0.878
Sampled PAM	4	0.462	0.308–0.662

The Consensus Agreement based on Average Linkage and Greedy mean NMI for both K-Means analyses at $k = 2$ and $k = 3$ was nearly identical. However, the generic NMF Ensemble Aggregation Method performed differently and was much less consistent than the other two methods: For example, a 2 Cluster K-Means Ensemble failed to converge after reaching an Iteration Limit, and collapsed into one cluster assignment. Similarly, the generic NMF Ensemble Aggregation Method used for the 3 Cluster K-Means Ensemble produced only two clusters. As such, this result relates only to the results of Ensemble Aggregation Methods, and should not be generalized to reflect performance of all possible implementations of the NMF Imputation Model.

4.12.3 Scaling, Missingness and Preprocessing Sensitivity

With robust scaling, discovery variances were 7.93 for language and 5.76 for orientation, compared with 0.44 for memory. Additionally, the first principal component of the data in the robust domain explained 63.3% of the total variability in the data and it was heavily loaded on both the language and the orientation dimensions. Narrow central ranges of ceiling-heavy scores amplify tails under interquartile-range scaling. With equal-domain scaling, the first principal component explained 58.5% of variance. We also saw substantial discrepancies between the three-group validation assignments made using the two types of scaling. In fact, we estimated an adjusted rand index (ARI) of .243 between the two sets of three-group validation assignments.

The average silhouette value for robust three-group labels was .608 in robust space and .440 in equal-domain space. In contrast, the average silhouette values for equal-domain labels were .172 and .283 for robust and equal-domain spaces respectively. Figure 4.14 separates the contributions from partitioning from those due to measurement geometry, neither geometry having been clinically validated. By comparison, application of equal-domain scaling resulted in lower average agreement levels among all possible combinations of imputed labels for two-group K-Means (.769), and for three-group K-Means (.747).

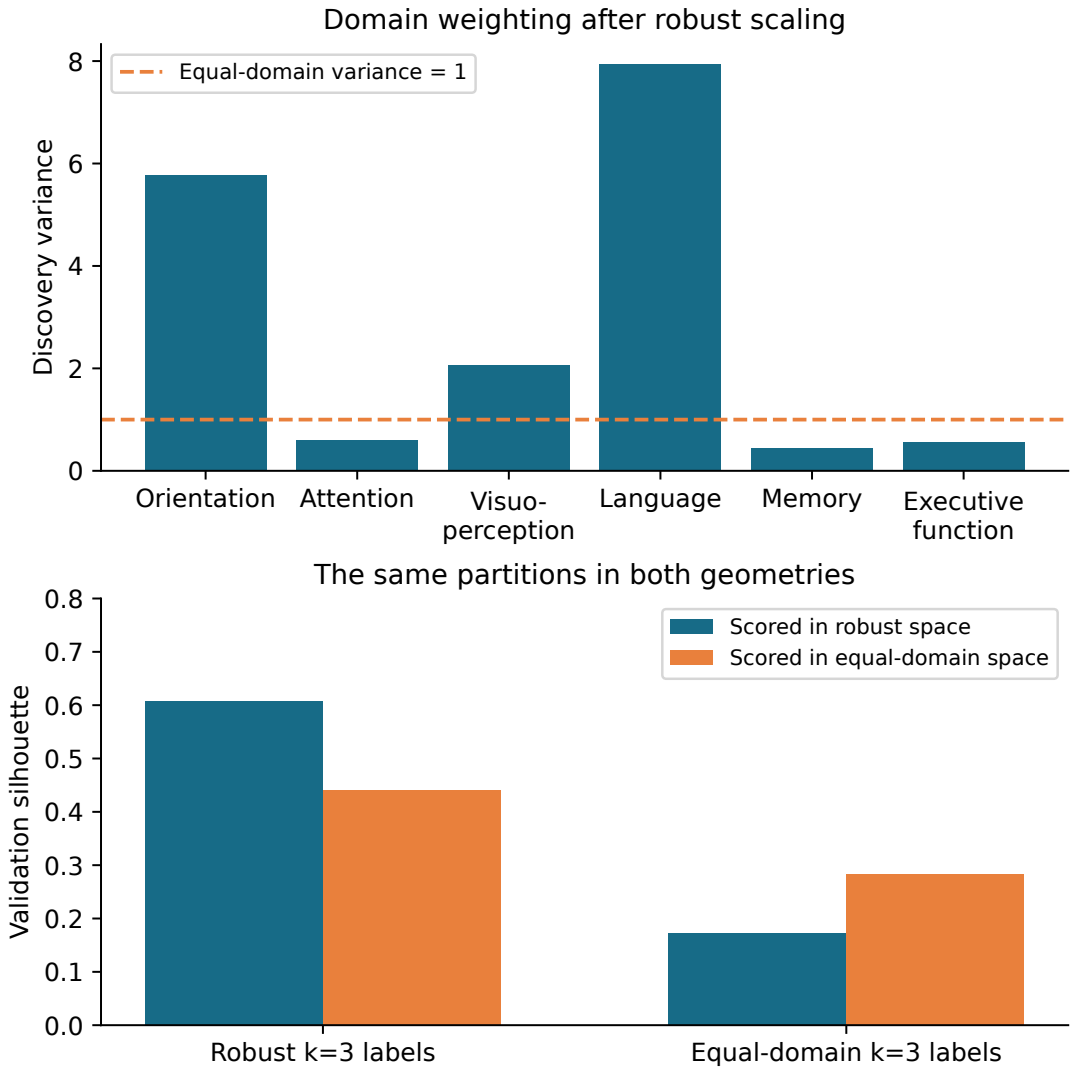


FIGURE 4.14: Scaling sensitivity. Top: domain variances after robust scaling, with unit-variance scaling as a reference. Bottom: the same two three-group partitions evaluated in both geometries. A high silhouette conditional on one geometry does not establish representation-independent subgroups.

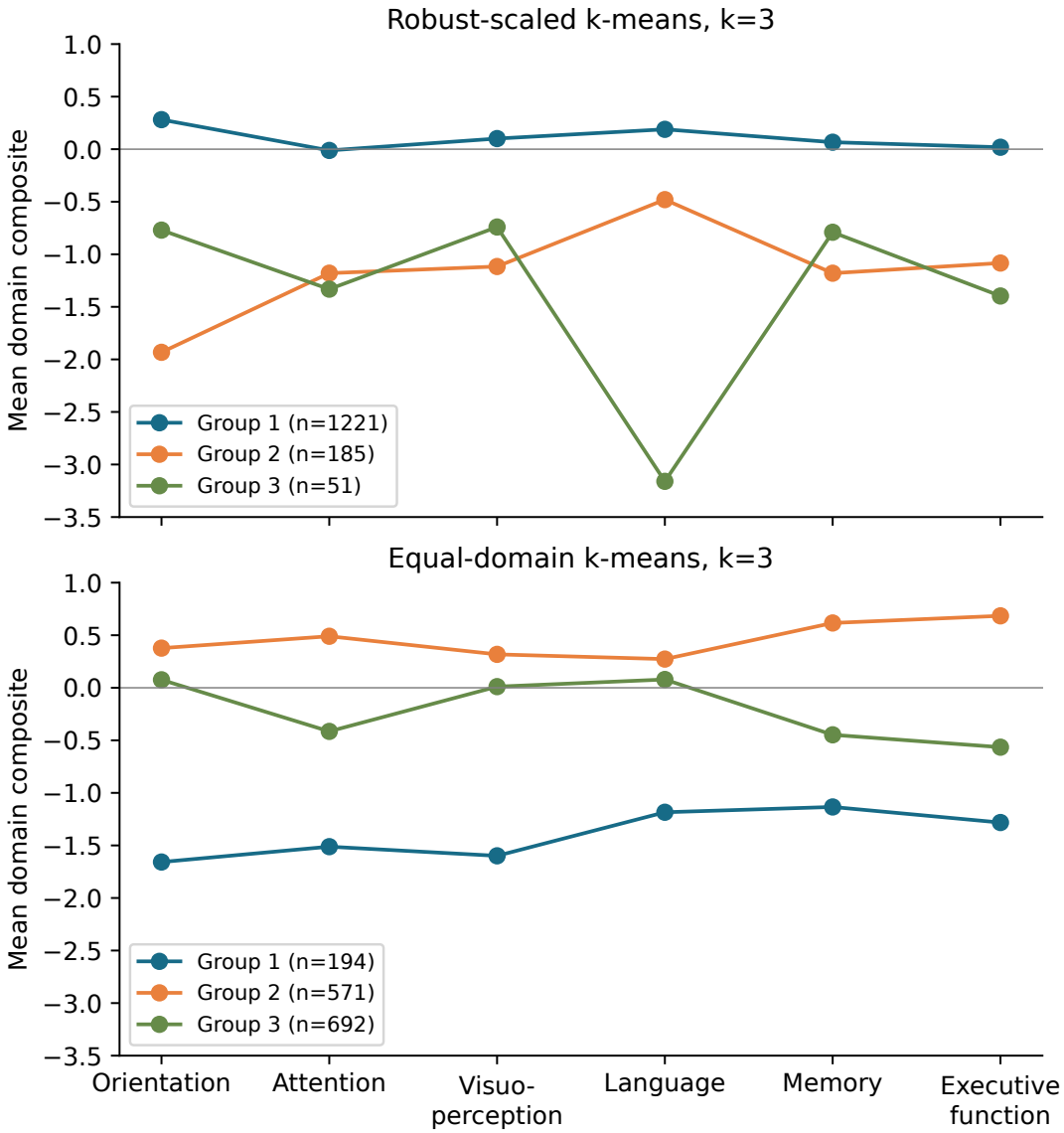


FIGURE 4.15: Validation profiles for fixed three-group k-means under robust and equal-domain scaling. Both panels show the same underlying direction-aligned domain composites. Group numbers and colors are local to each model; matching numbers across panels do not identify the same patients.

Robust three-group validation labels gave groups of 1221 patients, 185 patients, and 51 patients. The smallest group displayed clear deficits in terms of language abilities. Equal-domain scaling gave groups of 194 patients, 571 patients, and 692 patients with varying profiles compared to one another (see Figure 4.15). There is no empirical support here for transferring historical clinical tier names to the new labels.

There were also relations between the availability of testing and the fixed cluster allocations. Cramér's V for cluster allocation versus missingness was 0.412 for `MRAVLT075`, and 0.371 for `FEPMR`. Both tests had substantial variation in missingness across clusters, ranging from 10.2–80.4%, and 16.1–86.3%. Although this suggests some type of relation between cluster allocations and missingness, it does not provide enough evidence to suggest that missingness is what caused the clustering of the data. However, because of the variability present we cannot treat the reliability estimates generated using just one of the imputation techniques as evidence of no effect due to test administration.

The mean-imputed data set at $k = 3$ had an Adjusted Rand Index (ARI) of .921 compared to the reference grouping, while the k-nearest neighbors (KNN) imputed data set at $k = 3$ had an ARI of .538 as compared to the reference groupings. When we required at least seven observed tests per patient, 5,263 discovery and 1,339 validation patients remained. The ARI on common validation patients was 0.903. After de-clipping, we had an ARI of .958. Deterministically imputing with a 50 iteration limit, we achieved convergence by iteration #37, and had no label changes (ARI = 1.00). Using the "frozen" discovery transformation to cluster the validation patients independently, we obtained an ARI of .895. This statistic measures agreement between partitions; it is not another measure of predictive accuracy.

Comparing the results from the unaligned raw-test aggregation to those from the directionally aligned standardized-domain for the three-group partition, we found ARIs of .703 for all assessments, and .629 for complete-cases. There was also a high negative correlation (-0.749) between our two constructed attention-composite variables, so we suspect the scale/direction alignment affected these statistics. Thus, they are best viewed as sensitivity analyses using completed values as constants, and should not be considered repeats of our previously presented supplemental multiple imputation and complete-case analyses.

4.12.4 Incomplete-Data Mixtures and Continuous References

Of the 32 incomplete-data mixture fits, 28 achieved numerical convergence (Table 4.5). Furthermore, among all those which fulfilled both size and convergence criteria the t family with four degrees of freedom had the smallest BIC and highest validation likelihood. Both t-families selected eight components at the upper search bound, leaving the optimal component count unresolved. Their six-domain silhouettes were small. Likewise, the ICL identified eight components for the t families but only six rather than seven for the contaminated Gaussian.

TABLE 4.5: Selected observed-data mixtures fitted to 14 incomplete tests. Log likelihood is the mean per validation patient. Silhouette evaluates their assignments in the completed robust six-domain space.

Family	k	BIC	Validation log likelihood	Silhouette
Gaussian	6	73,878.4	−6.076	0.188
Contaminated Gaussian	7	74,641.8	−6.030	0.154
t, df = 10	8	69,815.8	−5.550	0.111
t, df = 4	8	67,545.6	−5.341	0.095

Although a better fit of a distribution to the data does not provide conclusive evidence regarding identifying eight distinct clinical phenotypes from the selected data it is possible that the multiple components identified represent either discrete score masses, skewness, and/or the tails of distributions.

Posterior probabilities quantify membership uncertainty conditional on the selected family and component count; they do not account for family-selection uncertainty or the missingness process.

As indicated above, the maximum k-means silhouette obtained in our 1000 patient continuous reference screen was 0.676 and the fitted-reference tail fraction was 0.0099. The PC1 dip-statistic tail fraction was 0.9901. We identified two clear clusters by the positive control using a silhouette measure of 0.917. As such, we have demonstrated using silhouette measures alone that we have surpassed at least one particular copula reference; however, we find no indication of excessive multimodality when comparing our projection statistics against this reference. These descriptive reference comparisons do not constitute an omnibus test establishing clinical phenotypes or excluding every continuous explanation.

4.12.5 Longitudinal Findings and Overall Interpretation

The total number of patients in the eligible trajectory cohort was 2,013. Of these, 1,590 belonged to the discovery group and 423 to the validation group. Convergence occurred for all of the four fitted class count combinations (Table 4.6). However, adding classes improved BIC and validation likelihood. At least one subgroup in the models with multiple classes had an expected membership below five percent. Because of this, we applied the rule to force the selection of one continuous patient random effect trajectory class. Multiclass resampling was not triggered. As such, it cannot be concluded that all patients recovered uniformly. The nonlinear variation and presence of very small subgroups still have to be resolved.

In addition to providing further support for the idea that the same cognitive assessment could be used to replicate similar partitions, this study's ability to generalize beyond its own findings is somewhat limited because replicating results is

TABLE 4.6: Patient trajectory models. The 5% minimum expected discovery-class size is an analysis rule, not a validated clinical threshold.

Classes	BIC	Validation log likelihood	Smallest expected class
1	8,376.3	−2.598	100.0%
2	7,572.2	−2.403	3.9%
3	6,806.0	−2.290	2.7%
4	6,465.5	−2.212	1.3%

contingent upon how the participants' data are represented and imputed. Specifically, there are several ways in which participant representation can vary. For example, how participants are scaled into groups, as well as what level of missingness exists within their data, will influence whether participants are assigned to groups. Also, the choice of assumed distributional families for the observed variables will impact both the number of components or clusters found and how those components/clusters are defined. Further, our results indicate that distinct clinical phenotypes were not established. The lack of availability of an independent functional outcome measure limits testing for clinical utility.

Chapter 5

Discussion

5.1 Cognitive Structure and the Limits of Phenotype Claims

The historical assessment showed a broad cognitive-severity structure among individuals with acquired brain injury (ABI). Patient-level extensions showed consistent score groupings that varied based upon scale selection and imputation. These assessments, individually and collectively, did not provide evidence for distinct clinical phenotypes; conversely, there was no direct evidence suggesting that clinical phenotypes may be non-existent. To classify historical assessments in the "above-average", "near-normal", or "global-impairment" tiers, I looked at performance across the whole battery. While each tier has different levels of performance, each does not represent a unique shape of performance. All six areas of the assessments produced similar patterns of movement. As can be observed in Figure 4.5, the first principal component explained approximately 56% of the total variability in domain scores. Additionally, it loaded positively on every single one of the six areas at virtually equivalent magnitude. Based upon this information, I interpret the results as showing a pattern of severity-based stratification, as opposed to identifying specific cognitive patterns. Furthermore, the tiers were associated with clinical unit assignment (see §4.7), but this analysis did not show additional predictive value beyond the broader diagnostic grouping.

In addition to the findings demonstrating the extent to which variation in neuropsychological test performance reflects variation associated with the domain tested versus a general factor, the findings further illustrate the size of the proportion of the recoverable structure within this routine data collection cohort that is accounted for by a general severity dimension. While secondary variations related to domain specificity were observed, I found no evidence to indicate that they would be responsible for determining clusters. Therefore, based upon the findings of these analyses, there exists empirical support for both clinical experience (i.e., global impairment patterns as a method of separating patients prior to identifying dissociative patterns) [Maas et al., 2017], and the positive manifold of inter-test correlations commonly reported in neuropsychological studies [Lezak et al., 2012]. The results of the phenotypic analysis presented herein also indicate that the findings represent a

somewhat conservative estimate of the phenomenon being studied when contrasted with those of previous examinations. For example, even after applying robustness procedures (i.e., imputation-based sensitivity testing) overall severity emerged as a defining characteristic of subgroups before profile pattern characteristics were identified. Given that domain-based aggregation and density-based clustering occurs along manifolds that are characterized by the presence of a general factor accounting for most of the variance among tests, I do not attach significant importance to the absence of evidence indicating narrow dissociations, including attention-specific subgroups, which may exist but be undetectable due to their small magnitude or orthogonal nature relative to the primary severity gradient.

While this current research does not conflict with previous studies on the variety of dissociable cognitive profiles that arise from brain injury [García-Rudolph et al., 2021] in terms of whether or not such profiles do exist, the results provide a qualification of how these prior findings should be interpreted. The methodology used by García-Rudolph et al. differs from mine in several ways. First, the methodology I employed (UMAP + HDBSCAN) allows identification of irregular shapes and/or "noise" at low densities. Second, I used assessment resampling to assess cluster reproduction. Finally, I standardize, winsorize, and align all of my variable measures in a specific direction before creating composite variables. Each preprocessing decision is significant – especially given large variance in raw score measurements (such as subject's completion time for Trail Making Tests), creation of artificial axes due to non-cognitive data, or large amounts of missing data where the density estimates may be skewed. Therefore apparent differences in phenotypes due to pre-processing decisions may not represent real structure differences in neuropsychology. This is discussed further in Chapter 6.

The first hypothesis was only partially supported (Section 4.5). All ten imputation methodologies produced adequate levels of both silhouette and noise metric assessments. In other words, all ten of the imputation methodologies recovered ordered bands of severity. However, while overall core Jaccard varied across the ten methodologies from 0.509 to 0.630, individual matched-cluster comparisons varied from 0.383 to 0.633 and cluster identifications were identified between three and five. These results support ordered severity bands, but do not establish sharply defined, uniformly reproducible phenotypes. Phenotype boundaries can shift without disrupting the underlying severity banding order.

I suggest that the three-tiered model produced from the original MICE pipeline represent an interpretable discrete representation of a continuum. Therefore, it cannot be considered as empirical evidence of natural kinds. Tier numbering was based on configuration and typically ranged from three to five in terms of number of tiers, when all configurations are compared. The cluster count varied depending on both the imputation analysis and the threshold sensitivity analyses; however, the ordering according to degree of cognitive impairment was consistent across all analyses. A continuous summary measure or clearly labeled descriptive categories warrant

additional validation. However, at this time, there has been no evaluation of the relative clinical utility of these two alternatives.

5.2 Imputation Sensitivity Under a Fixed Partition

I believe Hypothesis 2 represents my biggest contribution to methodology but in a very focused way. First, I applied both UMAP and HDBSCAN for identification of the primary clusters. Second, I performed a separate fixed-reference diagnostic test by performing one k-means fit to the MICE embedding and then projecting all of the other imputation data onto this fixed structure. In the diagnostic (see §4.6), the average ARI value was above 0.70 among the 45 method pairings and each of the 45 had an ARI greater than 0.50. The groupings were generally consistent across the imputed data in that fixed structure. In fact, I think the ability to identify the groupings is indicative of a strong severity gradient while multidimensional profile structures are harder to find and may be poorly defined or at least fragile. The two aspects of the grouping analysis could contribute to its robustness while allowing for variability in methods [Rubin, 1987].

Although I believe my method for dealing with the imputed data affected how many tiers I found, as well as their quality, I am inclined to think that my findings regarding the position of patients on the cognitive-severity continuum may vary less depending on what kind of imputation technique I employ. Overall, the fixed-reference diagnostic preserved many of the assignments between all compared imputations. The claim does not hold up when we examine independent model fits: the new KNN sensitivity yields an ARI of 0.538 against the reference three-group partition. That being said, while I do not feel these results indicate that missing data should be ignored, I would say that researchers can develop approaches to address missingness systematically and report on the methodology they use, and clinicians can treat tier labels as resources to support decision-making and judgment, rather than as replacements for educational, training and experiential based practice.

There was an obvious difference in the characteristics of both the smooth, model-based (EM, DAE, VAE) vs. the local/donor-type (KNN) imputation techniques (see Figure 4.9). While there is literature that categorizes various types of imputation methodologies (machine learning/statistical/deep learning), it seems to me that another way to categorize imputation methodologies is to separate them into two families: the "smooth" family of models where imputed values generally trend towards a model-based conditional mean, and local/donor models where estimates for a single assessment are made using a few neighboring assessments (donors).

The moderate association ($r = 0.47$; §4.6) identified by the correlation at the assessment level between the amount of imputed data and total instability suggests that subsequent analysis of the data will require separate consideration of assessments with larger amounts of missing information compared to assessments with relatively small amounts of missing information. This separate consideration may

take place through decreased confidence levels of assessments with high percentages of missing data or through multiple imputations being used to generate comparisons.

Consensus clustering offers an accessible way to address this challenge. Ten different imputation strategies were each given a "vote" on individual assessment labels. A consensus cluster vote produced a value representing how much agreement existed regarding assignment of labels to individual assessments based upon the voting results of the different imputation strategies. Across assessments, there was a high degree of average agreement (mean confidence = 0.93); moreover, 86.2% of assessments had high confidence. I find assessments with low levels of confidence to reflect clinical importance and not just inconvenience. These assessments may represent patients near tier boundaries and should prompt caution or reassessment.

Although MICE is generally accepted to replace missing data prior to clustering on neurocognitive test data, if a researcher anticipates that relationships between neuropsychological assessment variables will likely be nonlinear, MissForest [Stekhoven and Bühlmann, 2012] has been shown to perform well. Due to application of the Holm-Bonferroni adjustment procedure to each of the 45 pairs of comparisons (as described in Section 4.10), 44 of the 45 pairs rejected the null hypothesis stating that ARI ≤ 0.50. The exception was KNN–SoftImpute, whose ARI remained near the threshold. Therefore, within the fixed-reference H2 design, I concluded that there does not appear to be a single strategy for replacing missing data that drives patient placement on the severity dimension. The number of density-based tiers remains more sensitive when UMAP + HDBSCAN is refitted independently.

The amount of sensitivity possibly present in the treatment of missing data is illustrated in Section 4.10. Comparison of complete case data resulted in differences in the number of clusters identified, where silhouette = 0.257 whereas ARI = 0.625 when comparing MICE-derived solutions to the shared assessments derived through raw test aggregation. While the controlled comparisons in the extension demonstrated some degree of sensitivity related to the directional and scaled alignment, previous list-wise statistical analyses do not represent domain standardized analyses.

5.3 Clinical Unit Association

In order to assess if there is an association between cognitive-severity tier and which clinical unit each participant was assigned to, I used a contingency test on both tier and unit. There was a statistically significant association, with a small effect size according to Cohen's descriptive standards. This association was still present after controlling for the broader diagnostic category that was also provided through the clinical-unit lookup using a stratified sensitivity analysis. Since the broader category and unit details were both retrieved via the same clinical-unit lookup function, I did not view these two sources as providing independent validation.

Although the association may have some clinical implications, it cannot be taken as evidence of predictive capability or causality. When conducting a patient-grouped multinomial sensitivity analysis, when tier was added to the broad diagnostic grouping, held-out log loss decreased from 0.6064 to 0.6051. However, in assessing the patient bootstrap improvement interval for tier, zero was included; and accuracy was identical to three decimal points. Therefore, H3 provides only support for differences in tier distribution among units. Assessing prospective routing or future outcomes requires separate predictors, patient-grouped validation and clinical outcomes collected for that purpose.

5.4 Feature Engineering

Many ways exist to describe cognitive characteristics. Two general approaches use variable-level characteristics and aggregated characteristics. Variable-level characteristics evaluate each individual test by itself. Aggregated characteristics are averages or sums of tests and are designed to represent a specific cognitive domain. In my research, I tested whether an aggregated characteristic of a cognitive domain would have lower multicollinearity than a variable level characteristic. Better conditioning was assessed through lower variance inflation factors (VIF) and a lower condition number.

The results from my data analysis supported H4. My findings did demonstrate lower levels of multicollinearity when using domain-aggregated characteristics instead of variable level characteristics. More specifically, while I found that there were minimal differences in cluster separation (the silhouette value increased only slightly from 0.473 to 0.481), the average VIF decreased from 2.77 to 1.88 and the condition number decreased from 61.7 to 11.8 (see Section 4.8). Thus, I believe H4 provides evidence that aggregating domain-test scores can reduce redundancy and create a more interpretable and well-conditioned way of representing assessment data. It should be noted however that this finding does not provide evidence of materially stronger cluster separation. This advantage is also consistent with the practical use of assessing scores on a per-domain basis rather than examining them individually [Lezak et al., 2012].

I wanted to point out that there is a crucial requirement when you are combining test variables at the level of individual variables — all transformations at the variable level need to occur before the final step of aggregation. The data needs to undergo transformation (winsorization), alignment based on direction, and scale (standardization) before values from each variable can be put together into a single measure of overall performance or a composite. If this order is violated (i.e., if we aggregate values from different variables before standardizing those values), our results will artificially depend upon which variable has the greatest range because of its initial scale.

For instance, if several memory tests were aggregated prior to standardizing them, the final composite is likely to reflect the performance on the test(s) with the largest range rather than whatever aspect of memory it was intended to measure. Similarly, if multiple tests with a high degree of similarity (correlation) enter an unstandardized composite, UMAP and HDBSCAN may utilize the same dimension of cognition to inflate the distance computations.

Aggregating features produces a single feature per domain but does not necessarily ensure that each domain receives equivalent weighting after additional scaling. Additionally, as demonstrated by the patient-level extension, robust scaling provides a larger weight to the tails of both language and orientation.

5.5 Feature Importance and What Drives Severity

The results in §4.9 describe relationships among the cognitive domains and the classifications of the random forest surrogate model. Since the random forest surrogate model was 99.4% accurate in reproducing labels from the same domain scores used to create them, I believe that the ordering generated from the random forest surrogate primarily represents where the random forest surrogate model identified the boundaries of each partition generated during the fitting process rather than the causal reasons underlying each of the partitions. For example, since the random forest surrogate ranked orientation, visuoperceptual ability, and language as being higher than attention and executive function, I don't place much emphasis on the ordering. This is due to high correlations between many of the domains. Additionally, the Gini importance can be greatly influenced depending on whether you have multiple highly correlated predictors [Boulesteix et al., 2012].

I place more weight on the principal component analysis (PCA). The PCA indicated that only one principal component accounted for approximately 56 percent of the total variance in the domain scores. Each of the six domains had positive loadings that were all roughly equal in size. Therefore, I believe that this supports a general severity dimension rather than domain-specific dissociation or an attention specific pattern within this particular domain aggregated pipeline. The checks in §4.10 showed considerable assignment variability under assessment subsampling, and tier counts varied with variable-inclusion thresholds. The results demonstrate why researchers conducting data-driven research on phenotyping should explicitly document how their data was pre-processed and provide sensitivity analyses along with providing their original research findings.

5.6 Clinical Implications

I interpret this study's finding as evidence supporting a three-tiered cognitive severity stratification model. However, I see a lot of limitations to using such a model. As

I view it clinically, the most significant implication is conservative; they could provide some way of summarizing relative cognitive severity; but they cannot serve as independent clinical decision-making strategies. The medical education and knowledge of local service systems, as well as the professional experience of the clinician responsible for the referral to the rehabilitation services system, will continue to dictate how patients will be placed when they enter the rehabilitation services system. At this time, I do not believe that this research provides enough justification for direct clinical application (beyond carefully interpreting cognitive severity and openly dealing with missing data).

Historically, clinical practitioners have employed visual methods to examine distributions of patient scores in order to identify which patients represent "outliers" or "atypical" cases. Based upon what was found in this study, I also feel that there continues to be merit in employing algorithms to find clusters and visually examining the distribution of scores in order to identify outliers. Many of the assessments contained in the study had indistinct boundaries among separate groups. As such, I consider the middle tier of cognitive severity to be a part of a continuum with a boundary area cut out from it. Rather than falling into discrete categories of sharp separation, the majority of individuals appear to be distributed across a broad range of cognitive severity. As such, I believe that any future tools developed should focus on identifying assessments that either have difficulty clustering or are close to the boundary between two adjacent areas/clusters rather than attempting to assign each individual assessment a definitive phenotypic category.

Future research on large datasets that include missing cognitive assessment data may benefit from further statistical measures as well. One possible measure would be reporting the percentage of all assessments with an extremely low silhouette value (for example those with a silhouette value less than 0.05) in addition to both overall silhouette values and stability metrics. While this would still require substantive clinical interpretation by a professional, the inclusion of these percentages can provide a quantitative means of identifying the number of assessments located in areas where cluster membership is likely to be ambiguous.

5.7 What the Patient-Level Extension Adds

I investigated whether the patient groupings could be reproduced. In all twenty imputed datasets, two-group k-means clustering was identified as the most appropriate method to cluster the patients with a mean validation ARI of 0.959. Similarly, using three-group k-means, I found agreement across imputations in the original representation (ARI=0.935), and high agreement under patient subsampling. These analyses have extended the previous fixed-reference H2 diagnostic study by including separate re-fitting of partitions and maintaining the size of the validation set constant. Since the cohorts were generated differently and features were calculated

differently, I do not conclude that the increase in ARI from 0.710 to 0.935 represents an improvement over the historical result.

Repeating the assignment of patients to groups is essentially a different issue than validating the accuracy of a phenotype. The equal-domain scaling produced substantial changes in patient assignments based upon their performance on the various tests included in the cognitive battery (ARI = 0.243 versus robust-scaling) and reduced assignment agreement across imputations. Also, there were varying amounts of missing data in several of the memory and executive function tests for some groups. Regardless of how reliable a particular partition is, it may still be affected by either the geometric characteristics of the distribution of scores for the tests administered or other factors associated with how the tests were administered. One advantage of having a strong language-test contrast in one of the robust-space groups is its descriptive utility; however, a clinical interpretation would need to be evaluated under additional measurement and missingness models.

Likelihood results from these analyses also provide a further perspective on the comparisons. Although the restricted t mixtures were able to better fit the incomplete observations than did the selected Gaussian alternatives, as well as favoring eight components at the search boundary (as did the Gaussian models), neither family provided sufficient evidence to support the conclusion of a particular optimal number of patient types. Moreover, although there appeared to be some increase in the likelihood of trajectories with increasing numbers of classes, no multi-class solution could be identified that would satisfy the criteria for a minimal number of members in each class. Therefore, neither analysis can refute the possibility of the existence of very small subgroups.

These results address several issues associated with recently developed methodologies for dealing with incomplete data, ensembles, and longitudinal data [Tong and Tortora, 2025; Tomo et al., 2026; Julià et al., 2026], that employed many of the ideas presented in this study. Nevertheless, both the implementation and cohort differ from those in the referenced studies. Thus, this study supports the idea that reproducible cognitive score-based subgroupings exist when certain analytical conditions are met; however, this study does not provide support for clinically distinct phenotypes. It appears that a comparison using independent outcomes of continuous scores alone versus scores plus group membership, followed by external replication, would be required to determine whether identifying patients into pre-specified groups provides meaningful additional information.

Chapter 6

Limitations and Future Work

6.1 Limitations

Firstly, there is only one research site for this study; consequently, every piece of data collected in this study originated from a single rehabilitation unit. This may lead to selection bias which means that the ability to generalize these findings to other clinical populations cannot be confirmed. However, the tests are extensively studied and have international validity, while the six cognitive domains provide a common framework for describing post-injury cognition. Validation across multiple sites is required prior to considering this severity structure to represent ABI broadly instead of representing this specific rehabilitation unit.

Secondly, most historical analyses treated the 17,406 assessments as separate observations rather than grouping them within the 7,285 patients. While it is true that the repeated follow-up assessments of the same patient(s) will always be correlated by definition, neither the primary clustering method nor the assessment level bootstrap accounted for this correlation. Although utilizing each individual assessment allows for preserving the history of recovery status documented in the clinical records at each point in time, utilizing each individual assessment also can potentially create an overestimation of cluster stability along with creating narrower confidence intervals than what exists when assessments are grouped by patient. The patient-grouped multinomial sensitivity analysis directly assesses H3. The new extension adds additional techniques for both discovery and validation at the patient-level including patient subsampling and integrated random effects trajectory estimation. The new extension does not revise the previously calculated uncertainty intervals resulting from the historical assessment-level analysis.

Thirdly, the recovered model's structure imposes restrictions on how we are able to analyze our findings. Although the available historical data suggest that there exists a wide spectrum of possible degrees of severity, the extension demonstrates that while it is possible to establish replicable partitions amongst multiple ways of representing the same data, such partitions remain highly dependent upon which representation was used. Furthermore, it is not possible using the methodologies discussed herein to establish a singular categorical structure. A majority of the cohort exists along a single dimension that represents overall cognitive degree of

severity. The fact that PC1 accounted for approximately 56% of the total variance in the domain scores is evidence of this. The three tiers established through clustering represent a discretized version of this continuum. As such, the tiers should not be viewed as distinct, phenotypic entities with sharply defined boundaries. The actual number of tiers established will likely be influenced by two factors; namely, how one defines the threshold values separating the cluster(s) from one another and what proportion of the data is missing. Assessments that occur near the boundaries of each tier should also be approached with caution. A continuous severity score may be more representative than discrete tiers (Section 6.2).

Fourthly, the analytical methodologies have a limited ability to assess the temporal changes that occur post-injury. As previously mentioned, ABI recovery generally occurs very quickly in the initial phase and slowly thereafter [Ponsford et al., 2014; Maas et al., 2017]. Thus, a single evaluation represents a "snapshot" of an injured person's recovery process at a specific time. Many individuals had multiple evaluations in the dataset, allowing researchers to assess if they could detect patterns or trends of change; however, the timing of follow-up evaluations varied greatly as it depended on clinical needs rather than being part of a standardized protocol.

Of the total population of 5,206 patients who had longitudinal data available (Section 4.10), 50.2% remained in the same tier throughout all their study visits. The subset of 912 patients who had exactly two assessments and demonstrated a tier change showed a greater percentage of patients demonstrated increases in scores (n=706) compared to decreases (n=206); thus, the results are compatible with recovery, yet tier movement is not a functional outcome. There are additional variables that may impact tier movement including selection, discharge from care, timing of assessments, and availability of tests.

Further longitudinal studies with standardized assessment time frames are needed to address this question.

Another limitation is how missing values were imputed. As noted previously, it has been shown that the MAR assumption cannot be evaluated using only the observed data [Rubin, 1976; Little and Rubin, 2019]. For example, a patient may have a missing test score because he/she is unable to complete it. Therefore, if there is a Missing Not At Random (MNAR) mechanism for missingness, then the imputation of those missing values could overestimate test performance. Our comparisons do not provide an upper bound on this bias. We are planning to perform a formal sensitivity analysis of a "delta shift" to assess potential violations of our working assumptions regarding missingness.

Structured co-occurrence of missingness is consistent with test-battery protocols, but does not distinguish protocol-driven from outcome-dependent non-completion. Under a fixed MICE-fitted rule, at least 80% of the ten methods agreed on the assignment for 86.2% of assessments. This supports conditional assignment robustness, but does not establish ignorable missingness or agreement among independently fitted pipelines.

Assessment eligibility required at least one observed retained test, so records with many missing entries could be included. The higher number of required observations (seven tests) produced an index-patient population of 6,602. The adjusted Rand Index (ARI) value comparing these to the three-group assignments made by comparison on shared validation patients was 0.903. These results suggest some robustness in regards to excluding records based on sparseness; however, the status of the cognitive state of excluded encounters continues to remain uncertain.

The new observed-data mixture model addresses an issue with the previous historical method (impute-then-cluster) by integrating over missing coordinates. This new model continues to have assumptions about missing data (ignorable), utilizes continuous distributions as approximations for discrete and bounded score values, and uses a "dispersion-floor" to prevent singular solution(s). Fthe of the 32 attempts at fitting the models failed to meet the requested level of convergence. Both selected t-distribution families reached eight components, the upper end of the search. As such, the exact number of unique classes identified within these mixtures remain unknown. Additionally, I chose not to implement either skew-t or multivariate contaminated longitudinal models.

As such, feature engineering is still applicable. We remove all non-cognitive measures from the dataset. All test directions and scales are aligned. Winsorization is applied prior to creating composite scores through aggregation. Some potential artificial profile differences could arise based upon those decisions. The primary lesson regarding methodology is that unsupervised phenotyping relies heavily on preprocessing. Researchers will need to assess how variables were scaled, what variables were utilized in addition to how the study population was defined when assigning a clinical label to a distinct phenotype discovered using exploratory methods.

The use of HDBSCAN complicates the task of selecting the best approach to solving problems of this nature because, each parameter used in the models must be evaluated based on how sensitive the models are to those parameters. When we evaluate the performance of the model across all methodologies based on the composite metric Q; we are examining 9×120 possible permutations of the parameters. The biasing upward of the measure of separation obtained in the final embedded representation of the data with respect to an unbiased estimate of cluster quality occurs as a result of the incorporation into Q of both the silhouette value and the selection of the UMAP seed, which was selected at least partially based on the silhouette value. Based on results from PCA, there is evidence to suggest that there is one large factor (cognitive severity) associated with the data. This conclusion is further supported by the finding that 56% of the total variance can be explained by PC1. It should however also be acknowledged that there is considerable variation in the exact assignments of samples to groups based on how samples are grouped, as evidenced by a sub-sampling mean ARI of 0.273. Thus, these results do not identify a uniquely correct categorical partition.

Available computational resources were another constraint. While the process of

executing the UMAP + HDBSCAN procedure fifty times for each methodology was computationally intense; the modularity of the design enabled parallel processing (i.e., the procedures could be run concurrently), thus, although the cost of computer time for a typical consumer-level laptop was very high; it remained at acceptable levels.

Specific limitations existed in validating the extension. Feature selection was based on the historical full cohort. Equal-domain scaling was explored as a sensitivity after inspecting discovery variances. High ARI across bounded Bayesian imputations is contingent upon the imputation model, and it does not address MNAR. Test availability was associated with cluster membership; neither domain-weighting method has been clinically validated. The reference Gaussian copula was constructed using these same data and generates missingness independently of values. Its tail fractions are descriptive: they do not represent a calibrated omnibus test for overall significance. Although a positive control was run separately from the main analysis, this was not a power study, and no calibrated Silverman test was performed. Furthermore, the consensus summaries produced from validation-assignment vectors are not separately validated predictive models.

Finally, prior supplementary analyses comparing complete-case and multiply imputed datasets aggregated raw test values. In contrast, the patient-level extension aligned both the direction and scale of test results before aggregation. Controlled comparisons evaluated sensitivity to this difference; however, they did not recalculate all the previous statistics within the standardized-domain framework. Therefore, those previously calculated statistics cannot be used to validate the patient-level preprocessing pipeline.

6.2 Future Directions

Four prospective areas of research are identified based on the results and constraints discussed earlier.

One implication of the cognitive-severity gradients is that continuous indices of cognitive-severity (e.g. produced through PCA, or produced using supervised learning methods on metrics related to functional performance) can potentially provide better representations of patient status and can be used as better summaries of patient status than categorical tiers. Future studies should investigate how well continuous cognitive-severity indices perform relative to tiered classification schemes in terms of predicting prognosis and determining the optimal intensity of rehabilitation required based upon the needs of the individual, while acknowledging that tiered classification may represent an meaningful discrete representation of continuous data as opposed to representing inherent categories.

A second area of expansion would involve developing the linear random-effects mixture model for repeated assessments taken at fixed intervals, thereby allowing

investigators to understand longitudinal development of individuals' recovery trajectories. Specifically, the relationship between initial position along the cognitive-severity gradient at assessment and long term outcomes, as well as whether there are different rates of recovery associated with different levels of severity, could be investigated [Ponsford et al., 2014]. As previously mentioned, the selection of this longitudinal subset of trajectories was based on available visits and length of follow-up. Additionally, many other aspects remain unexamined including non-linear patterns of change, informative dropout, and additional externally derived measures of function. This current analysis has not established the value of these longitudinal subsets for either prognosis or rehabilitation planning.

Thirdly, multi-center validation is required for cognitive-severity stratification to be regarded as a feature of acquired brain injury (ABI) rather than this specific service. Multi-center studies that compare the outcomes from multiple health care systems and populations as well as different methodologies for assessing cognition will provide evidence of whether the one-dimensional cognitive-severity gradient described herein is consistent across populations with varied demographics, test batteries and schedules for assessing cognition [Maas et al., 2017].

Fourthly, including neuroimaging data will provide evidence of how the degree of cognitive impairment post ABI relates to underlying neurological mechanisms. Researchers will correlate an individual's position on the cognitive-severity gradient with measures of structural and functional imaging to determine if there are any neural substrates associated with global cognitive impairments after ABI. Such information may also provide guidance regarding the development of interventions [Saatman et al., 2008; Maas et al., 2017].

Chapter 7

Conclusions

7.1 Summary of Key Findings

The purpose of this research project was to determine if stable cognitive representations of acquired brain injury patients could be produced from the test data collected, even when a number of test scores are missing. Both the historical assessment analysis and the patient-level analysis indicated that similar patterns of assessment scores can be produced by consistently applying specific analytical options. However, neither demonstrated the existence of distinct clinical subgroups nor did they demonstrate the lack of such subgroups.

Three severity groupings were developed from 17,406 historical assessments via UMAP+HDBSCAN. Hypothesis H1 receives partial support. All ten imputation methods satisfied both the silhouette and noise criteria for an embedding. Moderate Jaccard agreement existed amongst the various imputation methods; however, the number of clusters differed across methods. A prominent first principal component, together with limited original-domain differentiation, suggests an overall severity interpretation within that representation.

Hypothesis H2 receives support within the fixed-reference design. Using the same MICE-fitted scaler and UMAP model and applying the three-group k-means rule to each of the ten imputed datasets resulted in an average pairwise adjusted Rand index (ARI) of .710. This represents sensitivity to imputation under predefined clustering rules; it does not represent independent discovery of density hierarchies. The historical assessment-level bootstrap ignores patient dependencies. Because the supplementary multiple imputation and complete-case analyses employed sums of raw test scores, their comparability with the standardized domain representations in the extension was limited.

H3 is supported as an association hypothesis (i.e., there is a relationship between clinical unit and tier; Cramér's V was 0.162). The addition of tier to broad diagnostic grouping resulted in a very small decline in patient-grouped validation log loss, however this improvement interval did include zero. H4 is primarily demonstrated through conditioning (i.e., domain aggregation reduced the mean VIF from 2.77 to 1.88 and the condition number from 61.7 to 11.8, while increasing silhouette slightly from 0.473 to 0.481).

7.2 Findings from the Patient-Level Extension

The use of an extension involving 7,285 index patients, divided into two groups (the discovery sample (n=5,828) and the validation sample (n=1,457)), selected two-group k-means in all twenty stochastic-imputation screens. The average pairwise validation ARI was 0.959. The overall ARI for the fixed three-group k-means clustering method averaged 0.935 over imputations and 0.968 over fifty random patient samples. Therefore, there appeared to be reproducible grouping patterns in the cognitive measures that were replicated across samples.

However, the issue of representation is the defining aspect of this problem. When using the equal-domain instead of the robust domain measure for scaling, the classifications produced by the three groupings of data (ARI = 0.243) changed substantially. In addition, after applying the robust domain scaling measure, language and orientation measures explained most of the variability in the data. Also, missing values on memory and executive function tests were found to relate to group membership. High levels of repeatability of the same preprocessing and imputation models do not necessarily translate to stable clinical categories.

The results from using restricted robust *t*-mixtures provided an improved fit to the data as compared to either of the two other Gaussian models. While both selected *t*-families had the same number of components and selected 8 components, the ambiguity in determining the appropriate number of components along with the relative lack of separation among the domains in the six-domain space suggests it may be unrealistic to assign eight unique phenotypic labels. Among the 2,013 patients eligible for longitudinal modelling, extra trajectory classes improved likelihood. However, the introduction of every multi-class trajectory solution resulted in a subgroup having less than the required 5% expected-membership threshold. Therefore, all that was left under the selection rules was a single class random effects model. Nonlinear variability and small groups remained unaddressed.

7.3 Methodological Contribution

Together, the workflows distinguish robustness under fixed-reference assignment from independent refitting, patient resampling, feature geometry and model fit to observed data. Shared preprocessing aligns test direction and scale before domain aggregation. Original-space metrics, posterior uncertainty, model convergence, conditional consensus and trajectory eligibility are reported explicitly. The implementation was checked numerically; clinical validation requires work beyond this study.

Beginning with the primary cohort, assessments were required to have a minimum of one retained test as the measure for determining whether an assessment contained sufficient test data. In addition to assessing the impact of missing data on clustering, a secondary sensitivity analysis examined the effects of limiting analyses to assessments that included seven or more observed tests to evaluate whether

eliminating sparse test profiles would also influence clustering. By making it clear the assessment-completeness rule, this method clearly defines the level of test data required to interpret stability.

7.4 Implications for Research and Clinical Interpretation

Continuous cognitive summaries and descriptive groups may remain candidates for future outcome validation. The current labeling system should include definitions used in preprocessing as well as any degree of uncertainty regarding those labels. These labels must not determine treatment intensities or routes for rehabilitation. Associations with clinical units or repeated cognitive assessments alone will not be sufficient to establish if treatments have produced clinically-relevant outcomes; however, CogDash presents both alternative designs for analyzing research and limitations for evaluating results and making comparisons among profiles from models that were saved during testing.

Empirical priorities for future research include: (1) comparing independently assessed outcome measurements based on continuous scores versus continuous scores with cluster membership; (2) examining the sensitivity of identified effects to missing data, especially where such data are informative; and (3) examining test-administration procedures. Furthermore, replicating the observed relationship among different sites at different time periods is also an empirical priority for future research. Nonlinear longitudinal modelling and the assessment of adequately measured small subgroups may serve as a foundation for future research that does not assume additional mixture components reflect legitimate patient types.

7.5 Final Remarks

The results of this study found that reproducible groupings could be established by examining cognitive scores. The membership in these groups and the total number of groups found can vary based upon the scale used, how missing data is handled and the choice of model type. While there was evidence in support of an overall structure of cognition, it also provided multiple pathways for additional research into subgroups of cognition. Distinct clinical phenotypes and their added clinical utility remain unconfirmed.

Bibliography

Afkanpour, Mitra et al. (2024). "Identify the most appropriate imputation method for handling missing values in clinical structured datasets: a systematic review". In: *BMC Medical Research Methodology* 24.1, p. 188.

Bellio, Marco et al. (2020). "Analyzing large Alzheimer's disease cognitive datasets: Considerations and challenges". In: *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring* 12.1. DOI: 10.1002/dad2.12135.

Boulesteix, Anne-Laure et al. (2012). "Overview of random forest methodology and practical guidance with emphasis on computational biology and bioinformatics". In: *WIREs Data Mining and Knowledge Discovery* 2.6, pp. 493–507. DOI: 10.1002/widm.1072.

Brazinova, Alexandra et al. (2021). "Epidemiology of Traumatic Brain Injury in Europe: A Living Systematic Review". In: *Journal of Neurotrauma* 38.10, pp. 1411–1440. DOI: 10.1089/neu.2015.4126.

Buuren, Stef van and Karin Groothuis-Oudshoorn (2011). "mice: Multivariate imputation by chained equations in R". In: *Journal of Statistical Software* 45.3, pp. 1–67. DOI: 10.18637/jss.v045.i03.

Chen, Bin et al. (Apr. 5, 2026). "Demystifying UMAP artifacts: An interactive study on diagnosis and steering using 3D probes". In: *Information Visualization* 25.3, pp. 358–372. DOI: 10.1177/14738716261434908. URL: https://journals.sagepub.com/doi/10.1177/14738716261434908 (visited on 10/15/2026).

Corrigan, John D, Anbesaw W Selassie, and Jean A Orman (2010). "The epidemiology of traumatic brain injury". In: *The Journal of Head Trauma Rehabilitation* 25.2, pp. 72–80. DOI: 10.1097/HTR.0b013e3181ccc8b4.

Dempster, Arthur P, Nan M Laird, and Donald B Rubin (1977). "Maximum likelihood from incomplete data via the EM algorithm". In: *Journal of the Royal Statistical Society: Series B* 39.1, pp. 1–38.

García-Rudolph, Alejandro et al. (2021). "Neuropsychological Assessments of Patients With Acquired Brain Injury: A Cluster Analysis Approach to Address Heterogeneity in Web-Based Cognitive Rehabilitation". In: *Frontiers in Neurology* 12, p. 701946.

Gondara, Lovedeep and Ke Wang (2018). "MIDA: Multiple Imputation using Denoising Autoencoders". In: *Advances in Knowledge Discovery and Data Mining (PAKDD)*. Springer, pp. 260–272.

Gravesteijn, Benjamin Y. et al. (2021). "Missing Data in Prediction Research: A Five-Step Approach for Multiple Imputation, Illustrated in the CENTER-TBI Study". In: *Journal of Neurotrauma* 38.13, pp. 1842–1857. DOI: 10.1089/neu.2020.7218.

Hennig, Christian (2007). "Cluster-wise assessment of cluster stability". In: *Computational Statistics & Data Analysis* 52.1, pp. 258–271.

Hubert, Lawrence and Phipps Arabie (1985). "Comparing partitions". In: *Journal of Classification* 2.1, pp. 193–218.

Julià, Laura et al. (May 23, 2026). "Clustering longitudinal data: comparison of model-based and distance-based approaches using simulated and real-world data in psychiatric research". In: *BMC Medical Research Methodology* 26, 159. DOI: 10.1186/s12874-026-02882-5. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC13374183/ (visited on 10/15/2026).

Kingma, Diederik P and Max Welling (2014). "Auto-encoding variational Bayes". In: *arXiv preprint arXiv:1312.6114*.

Lee, Daniel D and H Sebastian Seung (1999). "Learning the parts of objects by non-negative matrix factorization". In: *Nature* 401.6755, pp. 788–791.

Lezak, Muriel D et al. (2012). *Neuropsychological Assessment*. 5th ed. New York: Oxford University Press.

Little, Roderick JA and Donald B Rubin (2019). *Statistical Analysis with Missing Data*. 3rd ed. Hoboken, NJ: John Wiley & Sons. DOI: 10.1002/9781119482260.

Maas, Andrew IR et al. (2017). "Traumatic brain injury: integrated approaches to improve prevention, clinical care, and research". In: *The Lancet Neurology* 16.12, pp. 987–1048. DOI: 10.1016/S1474-4422(17)30371-X.

Mattei, Pierre-Alexandre and Jes Frellsen (2019). "MIWAE: Deep generative modelling and imputation of incomplete data sets". In: *Proceedings of the 36th International Conference on Machine Learning (ICML)*. Vol. 97, pp. 4413–4423.

Mazumder, Rahul, Trevor Hastie, and Robert Tibshirani (2010). "Spectral regularization algorithms for learning large incomplete matrices". In: *Journal of Machine Learning Research* 11, pp. 2287–2322.

McCoy, John T, Steve Kroon, and Lidia Auret (2018). "Variational autoencoders for missing data imputation with application to a simulated milling circuit". In: *IFAC-PapersOnLine* 51.21, pp. 141–146. DOI: 10.1016/j.ifacol.2018.09.406.

McInnes, Leland, John Healy, and James Melville (2018). "UMAP: Uniform Manifold Approximation and Projection for dimension reduction". In: *arXiv preprint arXiv:1802.03426*. DOI: 10.48550/arXiv.1802.03426.

Neville, Zachariah et al. (May 20, 2026). "A Comprehensive R Package for Clusterability Testing". In: *Journal of Open Research Software* 14, 37. DOI: 10.5334/jors.389. URL: https://reference-global.com/download/article/10.5334/jors.389.pdf (visited on 10/15/2026).

Pinsky, Eugene and Sidney Klawansky (2023). "MAD (about median) vs. quantile-based alternatives for classical standard deviation, skewness, and kurtosis". In:

Frontiers in Applied Mathematics and Statistics 9. DOI: 10.3389/fams.2023.1206537.

Ponsford, Jennie L et al. (2014). "Longitudinal follow-up of patients with traumatic brain injury: outcome at two, five, and ten years post-injury". In: *Journal of Neurotrauma* 31.1, pp. 64–77. DOI: 10.1089/neu.2013.2997.

Prakash, Prabhat et al. (2024). "Benchmarking Machine Learning Missing Data Imputation Methods in Large-Scale Mental Health Survey Databases". In: *medRxiv*. DOI: 10.1101/2024.05.13.24307231.

Rabinowitz, Amanda R and Harvey S Levin (2014). "Cognitive Sequelae of Traumatic Brain Injury". In: *Psychiatric Clinics of North America* 37.1, pp. 1–11. DOI: 10.1016/j.psc.2013.11.004.

Robert, Valérie, Yann Vasseur, and Vincent Brault (2020). "Comparing High-Dimensional Partitions with the Co-clustering Adjusted Rand Index". In: *Journal of Classification* 38.1, pp. 158–186. DOI: 10.1007/s00357-020-09379-w.

Rousseeuw, Peter J (1987). "Silhouettes: a graphical aid to the interpretation and validation of cluster analysis". In: *Journal of Computational and Applied Mathematics* 20, pp. 53–65.

Rubin, Donald B (1976). "Inference and missing data". In: *Biometrika* 63.3, pp. 581–592. DOI: 10.1093/biomet/63.3.581.

— (1987). *Multiple Imputation for Nonresponse in Surveys*. New York: John Wiley & Sons. DOI: 10.1002/9780470316696.

Saatman, Kathryn E et al. (2008). "Classification of traumatic brain injury for targeted therapies". In: *Journal of Neurotrauma* 25.7, pp. 719–738. DOI: 10.1089/neu.2008.0586.

Shannon, Claude E. (1948). "A Mathematical Theory of Communication". In: *Bell System Technical Journal* 27.3, pp. 379–423. DOI: 10.1002/j.1538-7305.1948.tb01338.x.

Stekhoven, Daniel J and Peter Bühlmann (2012). "MissForest—non-parametric missing value imputation for mixed-type data". In: *Bioinformatics* 28.1, pp. 112–118. DOI: 10.1093/bioinformatics/btr597.

Tomo, Yui, Funato Sato, and Mari Oba (June 1, 2026). "A comparative simulation study of cluster ensemble algorithms integrated with multiple imputation for clustering with missing data". In: *BMC Medical Research Methodology* 26, 168. DOI: 10.1186/s12874-026-02869-2. URL: https://link.springer.com/article/10.1186/s12874-026-02869-2 (visited on 10/15/2026).

Tong, Hung and Cristina Tortora (Dec. 19, 2025). "MixtureMissing: An R Package for Robust and Flexible Model-Based Clustering with Incomplete Data". In: *Journal of Statistical Software* 115.3, pp. 1–32. DOI: 10.18637/jss.v115.i03. URL: https://www.jstatsoft.org/article/view/v115i03 (visited on 10/15/2026).

Troyanskaya, Olga et al. (2001). "Missing value estimation methods for DNA microarrays". In: *Bioinformatics* 17.6, pp. 520–525.

UK Department of Health and Social Care (2022). *Acquired Brain Injury Call for Evidence*. URL: https://www.gov.uk/government/calls-for-evidence/acquired-brain-injury-call-for-evidence/acquired-brain-injury-call-for-evidence (visited on 06/21/2026).

Yoon, Jinsung, James Jordon, and Mihaela van der Schaar (2018). "GAIN: Missing data imputation using generative adversarial nets". In: *Proceedings of the 35th International Conference on Machine Learning (ICML)*. Vol. 80, pp. 5689–5698.

Appendix A

Imputation Hyperparameters

This appendix lists the hyperparameters used for each of the ten imputation methods evaluated in this thesis. Settings followed published defaults where defaults exist; remaining choices were selected by cross-validation or grid search as noted. The methods themselves are described in Chapter 3, ğ3.3.

TABLE A.1: Hyperparameters for the ten imputation methods.

Method	**Hyperparameters**
Mean	Single-pass arithmetic mean per variable.
EM	Multivariate normal model; convergence 10^{-6}; max 500 iterations.
MICE	Linear regression per variable; 10 iterations; single imputed dataset for clustering.
KNN	$k = 5$; inverse-distance weighting; Euclidean distance.
MissForest	100 trees per forest; up to 10 iterations; variables imputed in order of increasing missingness.
PMM	5 donors; 10 iterations; implemented within the MICE framework.
SoftImpute	Regularisation chosen by cross-validation over 20 log-spaced values from 10^{-3} to 10^{2}; convergence 10^{-5}; max 200 iterations.
NMF	Rank $r = 5$ selected by grid search over $\{2, 3, 5, 7, 10\}$; multiplicative updates; max 500 iterations.
DAE	Architecture $15 \to 128 \to 64 \to 32 \to 64 \to 128 \to 15$; ReLU + batch normalisation; dropout $(0.5, 0.3)$; corruption rate 20% with pattern-aware masking; MSE loss on observed entries; Adam optimiser (lr $= 10^{-3}$); ReduceLROnPlateau + early stopping (patience 20); 5 runs averaged.
VAE	Same layer dimensions as DAE; latent dim 32; loss = MSE + β-weighted KL divergence; cyclical annealing with $\beta_{\max} = 0.1$ over 40-epoch cycles; $K = 20$ latent samples averaged for final imputation.

Appendix B

List of Abbreviations

The abbreviations and acronyms used throughout this thesis are listed below.

ABI	Acquired Brain Injury
ANOVA	Analysis of Variance
ARI	Adjusted Rand Index (chance-corrected cluster agreement, -1 to $+1$)
BH	Benjamini–Hochberg (false-discovery-rate correction)
BIC	Bayesian Information Criterion (model-selection criterion)
CI	Confidence interval (95% unless otherwise stated)
CMH	Cochran-Mantel-Haenszel (stratified association test)
DAE	Denoising Autoencoder (deep-learning imputation)
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
ELBO	Evidence Lower Bound (VAE training objective)
EM	Expectation–Maximisation (maximum-likelihood imputation)
FDR	False discovery rate (controlled by Benjamini–Hochberg)
FWER	Family-wise error rate (controlled by Holm–Bonferroni)
GAIN	Generative Adversarial Imputation Networks
GAN	Generative Adversarial Network
GMM	Gaussian Mixture Model
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
IQR	Interquartile Range (spread used by the robust scaler)
KL	Kullback–Leibler (divergence term in the VAE loss)
KNN	K-Nearest Neighbours
KW	Kruskal-Wallis (non-parametric extension of one-way ANOVA)
MAE	Mean Absolute Error
MAR	Missing At Random
MCAR	Missing Completely At Random
MICE	Multiple Imputation by Chained Equations
MIDA	Multiple Imputation using Denoising Autoencoders
MLE	Maximum Likelihood Estimation
MNAR	Missing Not At Random
MSE	Mean Squared Error

NMF	Non-Negative Matrix Factorisation
NMI	Normalised Mutual Information (information-theoretic cluster agreement, 0 to 1)
NN	Near-Normal (the middle cognitive severity tier in the MICE solution)
PCA	Principal Component Analysis
PMM	Predictive Mean Matching (semi-parametric imputation, runs inside MICE)
RAVLT	Rey Auditory Verbal Learning Test
RMSE	Root Mean Squared Error
TBI	Traumatic Brain Injury
TMT	Trail Making Test
TSI	Time Since Injury (days at assessment)
t-SNE	t-distributed Stochastic Neighbour Embedding
UMAP	Uniform Manifold Approximation and Projection
VAE	Variational Autoencoder (deep-learning imputation with latent sampling)
VIF	Variance Inflation Factor (multicollinearity diagnostic)